

Figure 4 pAML reproducibility similarity analysis

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1 Env

```
rm(list = ls())  
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure4/")
```

2 Load libraries

```
library(FSA)
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
library(ineq)
library(grid)
library(vegan)
## Loading required package: permute
library(ArchR)
##
##
##                                     / |
##                               /      \
##                             .         !.
##                          \\          !.
##                        \\          `|.
##                       \\          |.
##                      \\          |\
##                     \\#####\       ||
##                    =====>        ||
##                   \|##==.....\     ||
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## \              '#-----,-,_,=##,_ ///
## , _--== _--,_,_-'#' ==' _-' | ##,-/
## -,___,---' \\####\_____,--\\_##,/
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## /__/_ \_\_\ | | \`-----| \_\_\_| | | | | | \`-----|
##
## ArchR : Version 1.0.3
## For more information see our website : www.ArchRProject.com
## If you encounter a bug please report : https://github.com/GreenleafLab/ArchR/issues
## Loading Required Packages...
## Loading Package : devtools v2.4.5
## Loading Package : grid v4.5.1
## Loading Package : gridExtra v2.3
## Loading Package : gtools v3.9.5
## Loading Package : gtable v0.3.6
## Loading Package : ggplot2 v4.0.0
## Loading Package : magrittr v2.0.4
## Loading Package : plyr v1.8.9
## Loading Package : stringr v1.5.2
## Loading Package : data.table v1.17.8
## Loading Package : matrixStats v1.5.0
## Loading Package : sparseMatrixStats v1.20.0
## Loading Package : S4Vectors v0.46.0
## Loading Package : GenomicRanges v1.60.0
## Loading Package : BiocGenerics v0.54.0
## Loading Package : Matrix v1.7.4
## Loading Package : Rcpp v1.1.0
## Loading Package : RcppArmadillo v15.0.2.2
```

```

## Loading Package : SummarizedExperiment v1.38.1
## Loading Package : rhdf5 v2.52.1
library(dplyr)
##
## Attaching package: 'dplyr'
## The following object is masked from 'package:Biobase':
##
##      combine
## The following objects are masked from 'package:GenomicRanges':
##
##      intersect, setdiff, union
## The following object is masked from 'package:GenomeInfoDb':
##
##      intersect
## The following objects are masked from 'package:IRanges':
##
##      collapse, desc, intersect, setdiff, slice, union
## The following objects are masked from 'package:S4Vectors':
##
##      first, intersect, rename, setdiff, setequal, union
## The following objects are masked from 'package:BiocGenerics':
##
##      combine, intersect, setdiff, setequal, union
## The following object is masked from 'package:generics':
##
##      explain
## The following object is masked from 'package:matrixStats':
##
##      count
## The following objects are masked from 'package:data.table':
##
##      between, first, last
## The following objects are masked from 'package:plyr':
##
##      arrange, count, desc, failwith, id, mutate, rename, summarise,
##      summarize
## The following object is masked from 'package:gridExtra':
##
##      combine
## The following objects are masked from 'package:stats':
##
##      filter, lag
## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union
library(GO.db)
## Loading required package: AnnotationDbi
##
## Attaching package: 'AnnotationDbi'
## The following object is masked from 'package:dplyr':
##
##      select
##

```

```

library(tidyr)
##
## Attaching package: 'tidyr'
## The following objects are masked from 'package:Matrix':
##
##     expand, pack, unpack
## The following object is masked from 'package:S4Vectors':
##
##     expand
## The following object is masked from 'package:magrittr':
##
##     extract
library(ggpubr)
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
##
## Attaching package: 'ggpubr'
## The following object is masked from 'package:plyr':
##
##     mutate
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
##
## Attaching package: 'sp'
## The following object is masked from 'package:IRanges':
##
##     %over%
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recommended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following object is masked from 'package:SummarizedExperiment':
##
##     Assays
## The following object is masked from 'package:GenomicRanges':
##
##     intersect
## The following object is masked from 'package:GenomeInfoDb':
##
##     intersect
## The following object is masked from 'package:IRanges':
##
##     intersect
## The following object is masked from 'package:S4Vectors':
##
##     intersect
## The following object is masked from 'package:BiocGenerics':
##
##     intersect

```

```

## The following objects are masked from 'package:base':
##
##      intersect, t
##
## Attaching package: 'Seurat'
## The following object is masked from 'package:SummarizedExperiment':
##
##      Assays
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##      method                from
##      heightDetails.titleGrobb ggplot2
##      widthDetails.titleGrobb  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##      as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##      annotate
library(stringr)
library(ggplot2)
library(circlize)
## =====
## circlize version 0.4.16
## CRAN page: https://cran.r-project.org/package=circlize
## Github page: https://github.com/jokergoo/circlize
## Documentation: https://jokergoo.github.io/circlize\_book/book/
##
## If you use it in published research, please cite:
## Gu, Z. circlize implements and enhances circular visualization
## in R. Bioinformatics 2014.
##
## This message can be suppressed by:
## suppressPackageStartupMessages(library(circlize))
## =====
library(pheatmap)
library(corrplot)
## corrplot 0.95 loaded
library(bedtoolsr)
## Warning in fun(libname, pkgname): bedtoolsr was built with bedtools version 2.30.0 but you have vers
## options(bedtools.path = "[bedtools path]")
library(tidyverse)
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr 2.1.5
## v lubridate 1.9.4    v tibble 3.3.0
## v purrr 1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%within%() masks IRanges::%within%()
## x ggpp::annotate() masks ggplot2::annotate()
## x dplyr::arrange() masks plyr::arrange()

```

```

## x dplyr::between()      masks data.table::between()
## x dplyr::collapse()    masks IRanges::collapse()
## x dplyr::combine()     masks Biobase::combine(), BiocGenerics::combine(), gridExtra::combine()
## x purrr::compact()     masks plyr::compact()
## x dplyr::count()       masks matrixStats::count(), plyr::count()
## x dplyr::desc()        masks IRanges::desc(), plyr::desc()
## x tidyr::expand()      masks Matrix::expand(), S4Vectors::expand()
## x tidyr::extract()     masks magrittr::extract()
## x dplyr::failwith()     masks plyr::failwith()
## x dplyr::filter()      masks stats::filter()
## x dplyr::first()       masks S4Vectors::first(), data.table::first()
## x lubridate::hour()    masks data.table::hour()
## x dplyr::id()          masks plyr::id()
## x lubridate::isoweek() masks data.table::isoweek()
## x dplyr::lag()         masks stats::lag()
## x dplyr::last()        masks data.table::last()
## x lubridate::mday()    masks data.table::mday()
## x lubridate::minute()  masks data.table::minute()
## x lubridate::month()   masks data.table::month()
## x ggpubr::mutate()     masks dplyr::mutate(), plyr::mutate()
## x tidyr::pack()        masks Matrix::pack()
## x BiocGenerics::Position() masks ggplot2::Position(), base::Position()
## x lubridate::quarter() masks data.table::quarter()
## x purrr::reduce()      masks GenomicRanges::reduce(), IRanges::reduce()
## x dplyr::rename()      masks S4Vectors::rename(), plyr::rename()
## x lubridate::second()  masks S4Vectors::second(), data.table::second()
## x lubridate::second<-() masks S4Vectors::second<-()
## x AnnotationDbi::select() masks dplyr::select()
## x purrr::set_names()   masks magrittr::set_names()
## x dplyr::slice()       masks IRanges::slice()
## x GenomicRanges::subtract() masks magrittr::subtract()
## x dplyr::summarise()    masks plyr::summarise()
## x dplyr::summarize()    masks plyr::summarize()
## x purrr::transpose()   masks data.table::transpose()
## x tidyr::unpack()      masks Matrix::unpack()
## x lubridate::wday()     masks data.table::wday()
## x lubridate::week()     masks data.table::week()
## x lubridate::yday()     masks data.table::yday()
## x lubridate::year()     masks data.table::year()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)
library(gridExtra)
library(data.table)
library(UpSetR)
library(ComplexUpset)
##
## Attaching package: 'ComplexUpset'
##
## The following object is masked from 'package:UpSetR':
##
##      upset
library(ggsankey)
library(RColorBrewer)

```

```

library(org.Hs.eg.db)
##
library(GenomicRanges)
library(ComplexHeatmap)
## =====
## ComplexHeatmap version 2.24.1
## Bioconductor page: http://bioconductor.org/packages/ComplexHeatmap/
## Github page: https://github.com/jokergoo/ComplexHeatmap
## Documentation: http://jokergoo.github.io/ComplexHeatmap-reference
##
## If you use it in published research, please cite either one:
## - Gu, Z. Complex Heatmap Visualization. iMeta 2022.
## - Gu, Z. Complex heatmaps reveal patterns and correlations in multidimensional
##   genomic data. Bioinformatics 2016.
##
##
## The new InteractiveComplexHeatmap package can directly export static
## complex heatmaps into an interactive Shiny app with zero effort. Have a try!
##
## This message can be suppressed by:
##   suppressPackageStartupMessages(library(ComplexHeatmap))
## =====
## ! pheatmap() has been masked by ComplexHeatmap::pheatmap(). Most of the arguments
##   in the original pheatmap() are identically supported in the new function. You
##   can still use the original function by explicitly calling pheatmap::pheatmap().
##
##
## Attaching package: 'ComplexHeatmap'
##
## The following object is masked from 'package:pheatmap':
##
##   pheatmap
library(clusterProfiler)
## clusterProfiler v4.16.0 Learn more at https://yulab-smu.top/contribution-knowledge-mining/
##
## Please cite:
##
## G Yu. Thirteen years of clusterProfiler. The Innovation. 2024,
## 5(6):100722
##
## Attaching package: 'clusterProfiler'
##
## The following object is masked from 'package:purrr':
##
##   simplify
##
## The following object is masked from 'package:AnnotationDbi':
##
##   select
##
## The following object is masked from 'package:IRanges':
##
##   slice

```

```
##
## The following object is masked from 'package:S4Vectors':
##
##     rename
##
## The following objects are masked from 'package:plyr':
##
##     arrange, mutate, rename, summarise
##
## The following object is masked from 'package:stats':
##
##     filter
library(networkD3)
##
## Attaching package: 'networkD3'
##
## The following object is masked from 'package:Seurat':
##
##     JS
##
## The following object is masked from 'package:SeuratObject':
##
##     JS
library(htmlwidgets)
##
## Attaching package: 'htmlwidgets'
##
## The following object is masked from 'package:networkD3':
##
##     JS
##
## The following object is masked from 'package:Seurat':
##
##     JS
##
## The following object is masked from 'package:SeuratObject':
##
##     JS
```

3 define functions

```
cosine_similarity <- function(a, b, na.rm = TRUE) {
  a <- as.numeric(a)
  b <- as.numeric(b)

  if (na.rm) {
    keep <- !(is.na(a) | is.na(b))
    a <- a[keep]
    b <- b[keep]
  }
}
```



```

denom <- sqrt(sum(a^2)) * sqrt(sum(b^2))
if (denom == 0) return(as.numeric(NA))

sim <- sum(a * b) / denom

sim <- max(min(sim, 1), -1)

return(as.numeric(sim))
}

retrieveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag", locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[, c("MutNum", "WTNum")]) > 0, ]
  cells_WT <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "WTNum"] > 0, ]$CellTag
  cells_Mut <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "MutNum"] > 0, ]$CellTag
  cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
  return(cells)
}

retrieveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList, ]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")]) > 0, ]
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0, ]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0, ]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}

get_mut_cols <- function(df, loc) {
  row <- df[df$Location == loc, ]
  mut_cols <- names(row)[row == "Mut"]
  mut_cols[mut_cols != "Location"]
}

```

AML_census.rds and *AML_tiergenes.txt* were downloaded from the COSMIC database and curated for downstream analyses

AML_SSM.rds and *AML_ATA_CSM_list.rds*, *AML_RNA_CSM_list.rds* were generated using scMutrace as described in the Methods section and *0_pipelineFunctions.Rmd* (pipelines for generating single-cell somatic mutation data list).

AML_scRNA_meta.rds and *AML_scATA_meta.rds* contain meta data for cell type annotation similarity analysis and were generate from *0_pipelineFunctions.Rmd* (pipelines for porcessing scRNA-seq data and

pipelines for processing scATAC-seq data).

sampleInfo.txt contains sample information including cell numbers and patient clinical data for scRNA-seq and scATAC-seq data and was downloaded from original paper.

EGA_scATA.hg38_multianno.txt and *EGA_scRNA.hg38_multianno.txt* contain mutation annotation information generated using ANNOVAR.

Mutational_Matrix_SBS.txt and *COSMIC_SBS96_Activities.txt* were generated using SigProfiler suite as described in the Methods section.

4 load data

```
# load AML census data
AML_census <- readRDS("../Data/HumanpAML/AML_census.rds")
AML_genes_tiers <- read.table("../Data/HumanpAML/AML_tiergenes.txt", header = T, sep = "\t")

# load SSM data
AML_SSM <- readRDS("../Data/HumanpAML/AML_SSM.rds")
AML_SSM_scRNA <- AML_SSM$scRNA
AML_SSM_scATA <- AML_SSM$scATA

# CSM
AML_ATA_CSM_list <- readRDS("../Data/HumanpAML/AML_ATA_CSM_list.rds")
AML_RNA_CSM_list <- readRDS("../Data/HumanpAML/AML_RNA_CSM_list.rds")

# meta data for celltype annotation similarity
AML_scRNA_meta_list <- readRDS("../Data/HumanpAML/AML_scRNA_meta.rds")
# metaQun
AML_scATA_meta_list <- readRDS("../Data/HumanpAML/AML_scATA_meta.rds")

AML_scRNA_meta <- do.call(rbind, AML_scRNA_meta_list)
rownames(AML_scRNA_meta) <- sub("\\.", "#", rownames(AML_scRNA_meta))

AML_scATA_meta <- do.call(rbind, AML_scATA_meta_list)
AML_scATA_meta$SampleID <- sub("_.*", "", AML_scATA_meta$SampleID)

rownames(AML_scATA_meta) <- sub("^AML[0-9]+\\.", "", rownames(AML_scATA_meta))

cellTypes <- unique(union(AML_scRNA_meta$Classified_Celltype, AML_scATA_meta$predictedGroup_Un))

# extract mutations and get bed format with census tag
AML_SSM_scRNA_bed <- separate(AML_SSM_scRNA, Location, into = c("chr", "pos", "ref", "alt"), sep = "_",
AML_SSM_scRNA_bed$end <- AML_SSM_scRNA_bed$pos
AML_SSM_scRNA_bed <- AML_SSM_scRNA_bed[,c("chr", "pos", "end", "ref", "alt", "Location")]
AML_SSM_scRNA_bed$Tag <- ifelse(AML_SSM_scRNA_bed$Location %in% AML_census$mutation$Location, "Census",
AML_scRNA_NotCensus_locs <- AML_SSM_scRNA_bed %>% dplyr::filter(Tag == "NotCensus") %>% pull(Location)

AML_SSM_scATA_bed <- separate(AML_SSM_scATA, Location, into = c("chr", "pos", "ref", "alt"), sep = "_",
AML_SSM_scATA_bed$end <- AML_SSM_scATA_bed$pos
AML_SSM_scATA_bed <- AML_SSM_scATA_bed[,c("chr", "pos", "end", "ref", "alt", "Location")]
AML_SSM_scATA_bed$Tag <- ifelse(AML_SSM_scATA_bed$Location %in% AML_census$mutation$Location, "Census",
AML_scATA_NotCensus_locs <- AML_SSM_scATA_bed %>% dplyr::filter(Tag == "NotCensus") %>% pull(Location)
```

```

# load paired sample info data
AML_sampleInfo <- read.table("../Data/HumanpAML/sampleInfo.txt", header = T, sep = "\t")
AML_sampleInfo[is.na(AML_sampleInfo)] <- 0

# AML ann
AML_scATA_ann <- read.table("../Data/HumanpAML/EGA_scATA.hg38_multianno.txt", header = T, sep = "\t")
AML_scrRNA_ann <- read.table("../Data/HumanpAML/EGA_scrRNA.hg38_multianno.txt", header = T, sep = "\t")

col_freq_databases <- c("gnomad40_exome_AF", "gnomad40_exome_AF_raw", "gnomad40_exome_AF_XX",
                        "gnomad40_exome_AF_XY", "gnomad40_exome_AF_grpmax", "gnomad40_exome_faf95",
                        "gnomad40_exome_faf99", "gnomad40_exome_fafmax_faf95_max", "gnomad40_exome_fafmax_faf99_max",
                        "gnomad40_exome_AF_afr", "gnomad40_exome_AF_amr", "gnomad40_exome_AF_asj",
                        "gnomad40_exome_AF_eas", "gnomad40_exome_AF_fin", "gnomad40_exome_AF_mid",
                        "gnomad40_exome_AF_nfe", "gnomad40_exome_AF_remaining", "gnomad40_exome_AF_sas",
                        "gnomad40_genome_AF", "gnomad40_genome_AF_raw", "gnomad40_genome_AF_XX",
                        "gnomad40_genome_AF_XY", "gnomad40_genome_AF_grpmax", "gnomad40_genome_faf95",
                        "gnomad40_genome_faf99", "gnomad40_genome_fafmax_faf95_max",
                        "gnomad40_genome_fafmax_faf99_max", "gnomad40_genome_AF_afr",
                        "gnomad40_genome_AF_ami", "gnomad40_genome_AF_amr",
                        "gnomad40_genome_AF_asj", "gnomad40_genome_AF_eas",
                        "gnomad40_genome_AF_fin", "gnomad40_genome_AF_mid",
                        "gnomad40_genome_AF_nfe", "gnomad40_genome_AF_remaining", "gnomad40_genome_AF_sas")

col_freq_databases %in% colnames(AML_scATA_ann)
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
col_freq_databases %in% colnames(AML_scrRNA_ann)
## [1] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [16] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
## [31] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE

AML_scATA_ann <- AML_scATA_ann %>%
  mutate(across(all_of(col_freq_databases),
    ~ as.numeric(replace(., . == ".", 0))))

AML_scrRNA_ann <- AML_scrRNA_ann %>%
  mutate(across(all_of(col_freq_databases),
    ~ as.numeric(replace(., . == ".", 0))))

# Annotate bed census and nonCensus
# scATA
AML_SSM_scATA_bed <- AML_SSM_scATA_bed %>%
  left_join(
    AML_scATA_ann %>%
      dplyr::select(Location, all_of(col_freq_databases)),
    by = "Location"
  ) %>%
  mutate(
    freq = if_else(
      if_any(all_of(col_freq_databases), ~ . > 0.0001),
      "high", "rare"
    )
  )

```

```

) %>%
dplyr::select(names(AML_SSM_scATA_bed), freq)

# scRNA
AML_SSM_scRNA_bed <- AML_SSM_scRNA_bed %>%
  left_join(
    AML_scRNA_ann %>%
      dplyr::select(Location, all_of(col_freq_databases)),
    by = "Location"
  ) %>%
  mutate(
    freq = if_else(
      if_any(all_of(col_freq_databases), ~ . > 0.0001),
      "high", "rare"
    )
  ) %>%
dplyr::select(names(AML_SSM_scRNA_bed), freq)

# load mutation features data
# generate using sigProfiler suite
AML_mutation_features <- read.table("../Data/HumanpAML/Mutational_Matrix_SBS.txt", header = T, sep = "\t")
AML_mutation_features_contribution <- read.table("../Data/HumanpAML/COSMIC_SBS96_Activities.txt", header = T, sep = "\t")

```

5 mutation Similarity DXREL for lineage switching (downsample)

```

AML_scRNA_CSM_list <- AML_RNA_CSM_list
AML_scATA_CSM_list <- AML_ATA_CSM_list

# scRNA
# AML_CSM_DX_list
AML_scRNA_CSM_DX_list <- lapply(AML_scRNA_CSM_list, function(df) {
  df[grepl("DX", rownames(df)), , drop = FALSE]
})
AML_scRNA_CSM_DX_list <- Filter(function(x) nrow(x) > 0, AML_scRNA_CSM_DX_list)

# AML_CSM_REM_list
AML_scRNA_CSM_REM_list <- lapply(AML_scRNA_CSM_list, function(df) {
  df[grepl("REM", rownames(df)), , drop = FALSE]
})
AML_scRNA_CSM_REM_list <- Filter(function(x) nrow(x) > 0, AML_scRNA_CSM_REM_list)

# AML_CSM_REL_list
AML_scRNA_CSM_REL_list <- lapply(AML_scRNA_CSM_list, function(df) {
  df[grepl("REL", rownames(df)), , drop = FALSE]
})
AML_scRNA_CSM_REL_list <- Filter(function(x) nrow(x) > 0, AML_scRNA_CSM_REL_list)

# scATA
# AML_CSM_DX_list
AML_scATA_CSM_DX_list <- lapply(AML_scATA_CSM_list, function(df) {
  df[grepl("DX", rownames(df)), , drop = FALSE]
})

```

```

})
AML_scATA_CSM_DX_list <- Filter(function(x) nrow(x) > 0, AML_scATA_CSM_DX_list)

# AML_CSM_REM_list
AML_scATA_CSM_REM_list <- lapply(AML_scATA_CSM_list, function(df) {
  df[grepl("REM", rownames(df)), , drop = FALSE]
})
AML_scATA_CSM_REM_list <- Filter(function(x) nrow(x) > 0, AML_scATA_CSM_REM_list)

# AML_CSM_REL_list
AML_scATA_CSM_REL_list <- lapply(AML_scATA_CSM_list, function(df) {
  df[grepl("REL", rownames(df)), , drop = FALSE]
})
AML_scATA_CSM_REL_list <- Filter(function(x) nrow(x) > 0, AML_scATA_CSM_REL_list)

HSPC <- c("HSC", "Progenitor")
Myeloid <- c("GMP", "Monocytes", "CD16.Monocytes", "cDC", "pDC")
Lymphoid <- c("CLP", "Pre.B.Cell", "B.Cell", "CD4.Naive", "CD4.Memory", "CD8.Naive", "CD8.Memory", "CD8")
Erythroid_Basophil <- c("Early.Erythrocyte", "Late.Erythrocyte")
Other <- c("Early.Basophil", "Plasma")

AML_scrNA_meta <- AML_scrNA_meta %>%
  dplyr::mutate(
    maincelltype = case_when(
      Classified_Celltype %in% HSPC ~ "HSPC",
      Classified_Celltype %in% Myeloid ~ "Myeloid",
      Classified_Celltype %in% Lymphoid ~ "Lymphoid",
      Classified_Celltype %in% Erythroid_Basophil ~ "Erythroid",
      Classified_Celltype %in% Other ~ "Other",
      TRUE ~ NA_character_
    )
  )

AML_scATA_meta <- AML_scATA_meta %>%
  dplyr::mutate(
    maincelltype = case_when(
      predictedGroup_Un %in% HSPC ~ "HSPC",
      predictedGroup_Un %in% Myeloid ~ "Myeloid",
      predictedGroup_Un %in% Lymphoid ~ "Lymphoid",
      predictedGroup_Un %in% Erythroid_Basophil ~ "Erythroid",
      predictedGroup_Un %in% Other ~ "Other",
      TRUE ~ NA_character_
    )
  )

AML_scATA_meta$Classified_Celltype <- AML_scATA_meta$predictedGroup_Un

maincelltypes <- c("HSPC", "Myeloid", "Lymphoid", "Erythroid", "Other")
subcelltypes <- c(HSPC, Myeloid, Lymphoid, Erythroid_Basophil, Other)

AML_scrNA_meta_DX <- AML_scrNA_meta %>%
  dplyr::filter(grepl("DX", Library_ID))

```

```

AML_scrRNA_meta_REL <- AML_scrRNA_meta %>%
  dplyr::filter(grepl("REL", Library_ID))

AML_scATA_meta_DX <- AML_scATA_meta %>%
  dplyr::filter(grepl("DX", Sample))

AML_scATA_meta_REL <- AML_scATA_meta %>%
  dplyr::filter(grepl("REL", Sample))

selected_samples_id <- c("AML4", "AML5", "AML6", "AML8", "AML10", "AML12", "AML16", "AML18", "AML21")
selected_samples_id <- c("AML2", "AML4", "AML5", "AML6", "AML8", "AML10", "AML11", "AML12",
  "AML13", "AML15", "AML16", "AML17", "AML18", "AML19", "AML21",
  "AML22", "AML24", "AML25", "AML26", "AML27")

AML_scrRNA_CSM_DX_select_list <- AML_scrRNA_CSM_DX_list[names(AML_scrRNA_CSM_DX_list) %in% selected_samples_id]
names(AML_scrRNA_CSM_DX_select_list) <- paste(selected_samples_id, "DX", sep = "_")
AML_scrRNA_CSM_REL_select_list <- AML_scrRNA_CSM_REL_list[names(AML_scrRNA_CSM_REL_list) %in% selected_samples_id]
names(AML_scrRNA_CSM_REL_select_list) <- paste(selected_samples_id, "REL", sep = "_")
AML_scrRNA_CSM_DX_REL_select_list <- c(AML_scrRNA_CSM_DX_select_list, AML_scrRNA_CSM_REL_select_list)

AML_scrRNA_meta_DX$Classified_CelltypeQ <- paste(AML_scrRNA_meta_DX$Lambo_et_al_ID, "DX", AML_scrRNA_meta_DX$Classified_CelltypeQ)
AML_scrRNA_meta_REL$Classified_CelltypeQ <- paste(AML_scrRNA_meta_REL$Lambo_et_al_ID, "REL", AML_scrRNA_meta_REL$Classified_CelltypeQ)
AML_scrRNA_meta_DX$maincelltypeQ <- paste(AML_scrRNA_meta_DX$Lambo_et_al_ID, "DX", AML_scrRNA_meta_DX$maincelltypeQ)
AML_scrRNA_meta_REL$maincelltypeQ <- paste(AML_scrRNA_meta_REL$Lambo_et_al_ID, "REL", AML_scrRNA_meta_REL$maincelltypeQ)

AML_scATA_CSM_DX_select_list <- AML_scATA_CSM_DX_list[names(AML_scATA_CSM_DX_list) %in% selected_samples_id]
names(AML_scATA_CSM_DX_select_list) <- paste(selected_samples_id, "DX", sep = "_")
AML_scATA_CSM_REL_select_list <- AML_scATA_CSM_REL_list[names(AML_scATA_CSM_REL_list) %in% selected_samples_id]
names(AML_scATA_CSM_REL_select_list) <- paste(selected_samples_id, "REL", sep = "_")
AML_scATA_CSM_DX_REL_select_list <- c(AML_scATA_CSM_DX_select_list, AML_scATA_CSM_REL_select_list)
AML_scATA_meta_DX$Classified_CelltypeQ <- paste(AML_scATA_meta_DX$SampleID, "DX", AML_scATA_meta_DX$Classified_CelltypeQ)
AML_scATA_meta_REL$Classified_CelltypeQ <- paste(AML_scATA_meta_REL$SampleID, "REL", AML_scATA_meta_REL$Classified_CelltypeQ)

AML_scrRNA_meta_DX_REL <- rbind(AML_scrRNA_meta_DX, AML_scrRNA_meta_REL)
AML_scATA_meta_DX_REL <- rbind(AML_scATA_meta_DX, AML_scATA_meta_REL)

# samples for lineage switching

AML_scrRNA_meta_DX_REL <- AML_scrRNA_meta_DX_REL[AML_scrRNA_meta_DX_REL$Lambo_et_al_ID %in% selected_samples_id,]
AML_scrRNA_meta_DX_REL$CellTag <- rownames(AML_scrRNA_meta_DX_REL) %>%
  sub("-1$", "", .) %>%
  gsub("#", "_", .)
AML_scrRNA_meta$CellTag <- rownames(AML_scrRNA_meta) %>%
  sub("-1$", "", .) %>%
  gsub("#", "_", .)

AML_scrRNA_meta_DX_REL$SampleID <- AML_scrRNA_meta_DX_REL$Lambo_et_al_ID
AML_scATA_meta_DX_REL <- AML_scATA_meta_DX_REL[AML_scATA_meta_DX_REL$SampleID %in% selected_samples_id,]
AML_scATA_meta_DX_REL$CellTag <- rownames(AML_scATA_meta_DX_REL) %>%
  sub("-1$", "", .) %>%
  gsub("#", "_", .)
AML_scATA_meta$CellTag <- rownames(AML_scATA_meta) %>%
  sub("-1$", "", .) %>%
  gsub("#", "_", .)

```

```

gsub("#", "_", .)

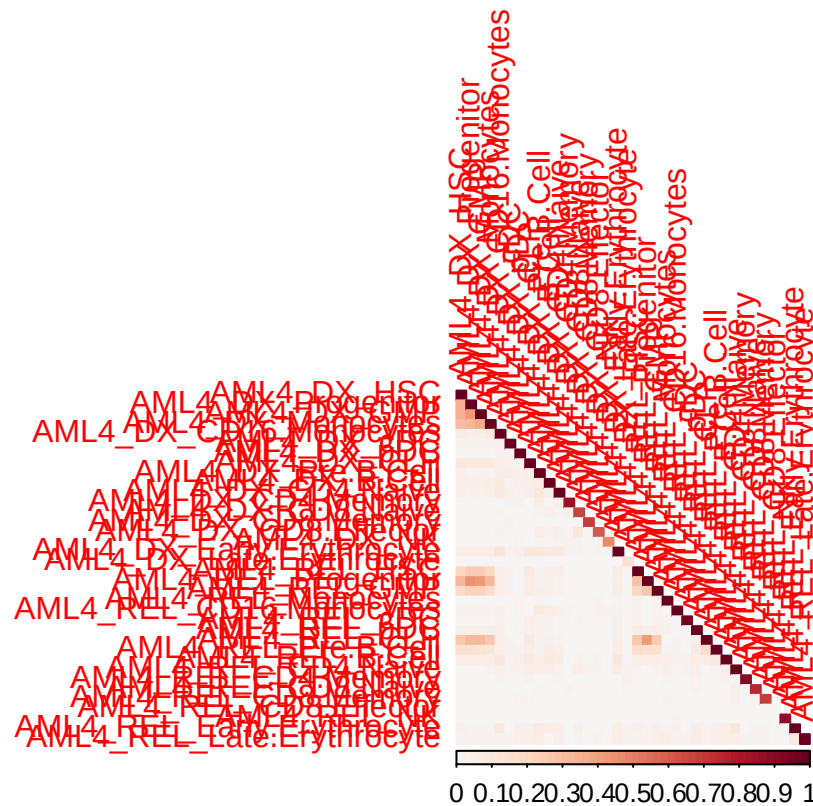
AML_scATA_meta_DX_REL$maincelltypeQ <- paste(AML_scATA_meta_DX_REL$Sample, AML_scATA_meta_DX_REL$maince
cells_limits <- 0

celltypes_scRNA <- names(table(AML_scRNA_meta_DX_REL$Classified_CelltypeQ)[table(AML_scRNA_meta_DX_REL$
celltypes_scATA <- names(table(AML_scATA_meta_DX_REL$Classified_CelltypeQ)[table(AML_scATA_meta_DX_REL$

celltypes_scRNA_main <- names(table(AML_scRNA_meta_DX_REL$maincelltypeQ)[table(AML_scRNA_meta_DX_REL$ma
celltypes_scATA_main <- names(table(AML_scATA_meta_DX_REL$maincelltypeQ)[table(AML_scATA_meta_DX_REL$ma

```

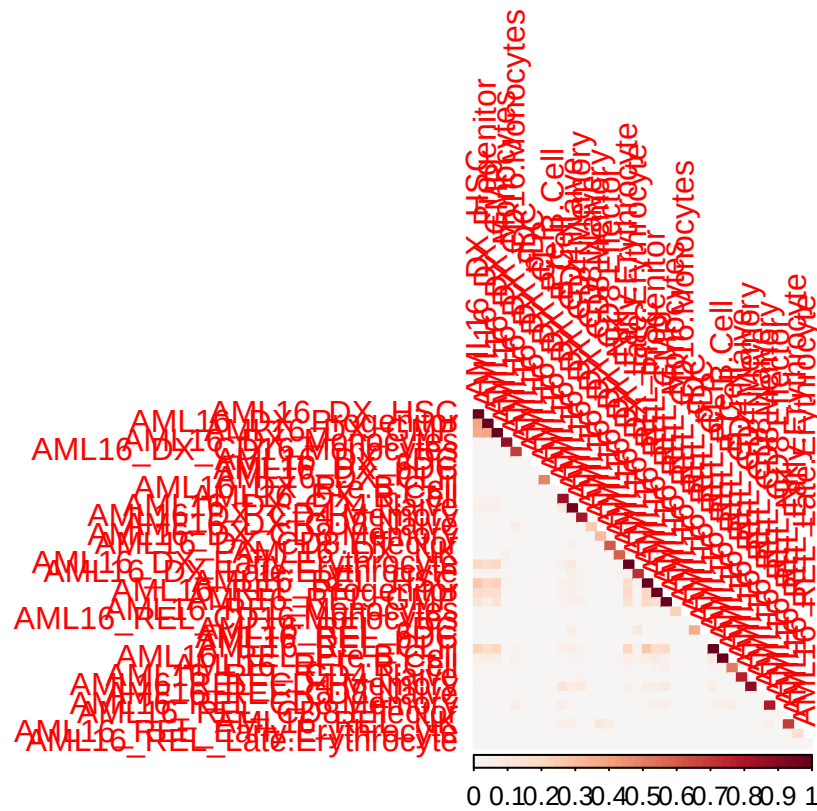
6 calSimilarity downsampling



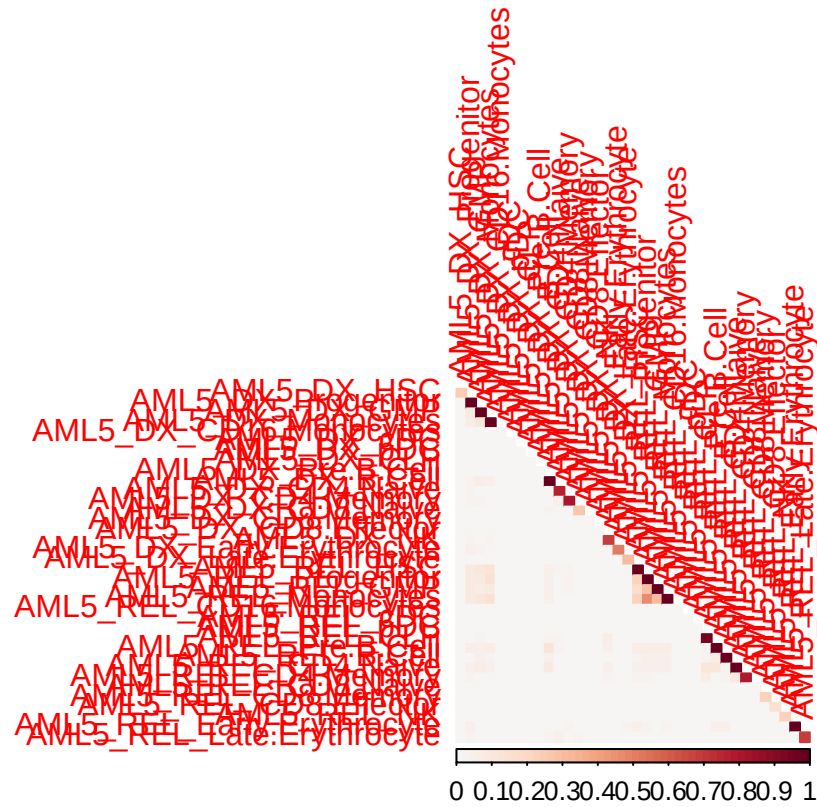
```

corrplot(mat_sorted_AML16_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o

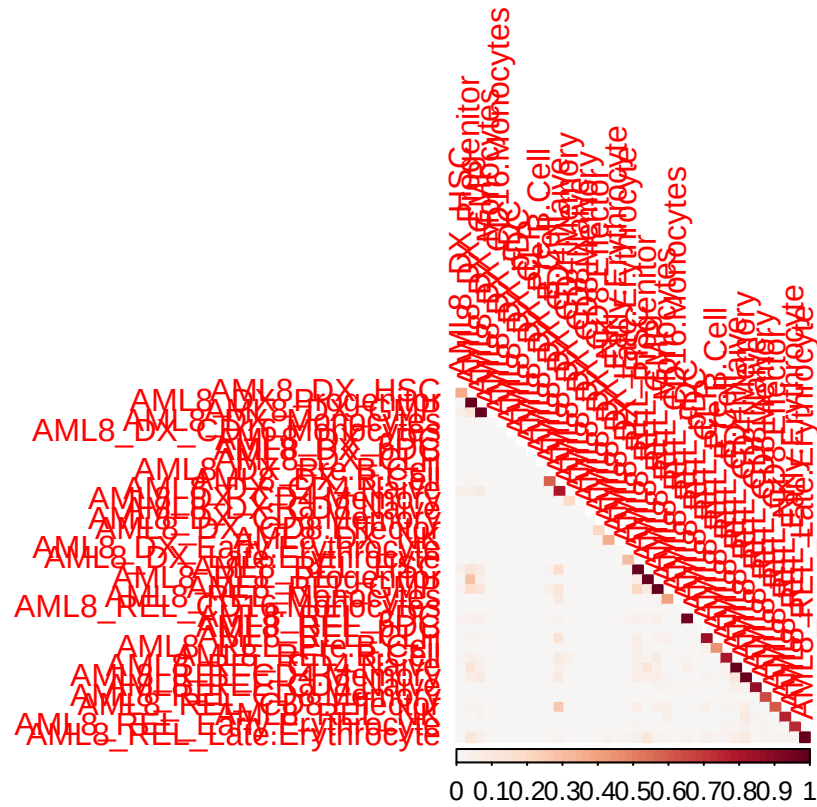
```



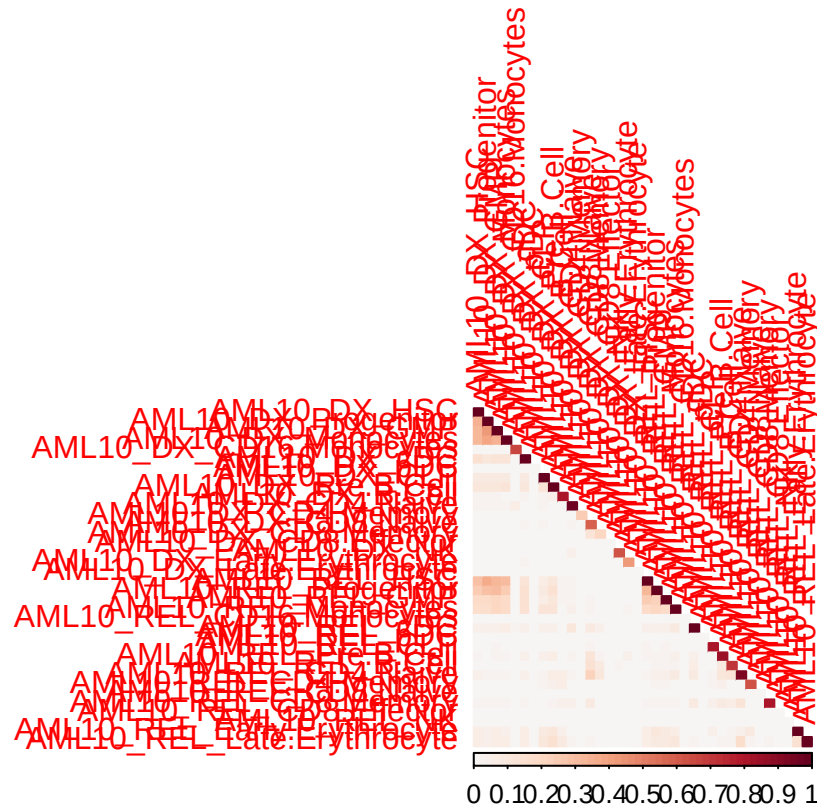
```
corrplot(mat_sorted_AML05_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```

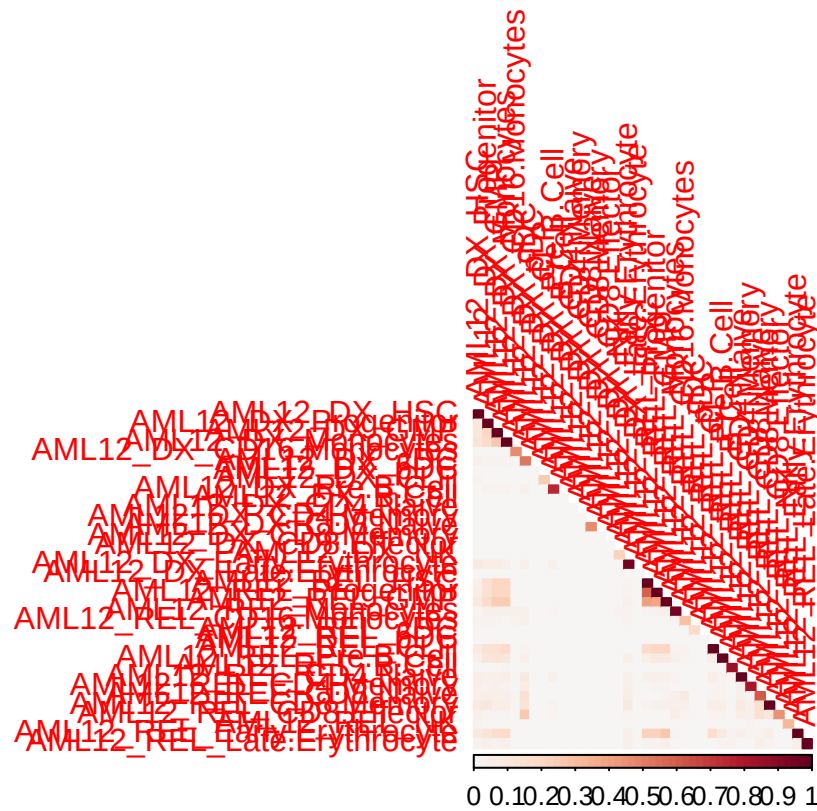
```
corrplot(mat_sorted_AML06_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```

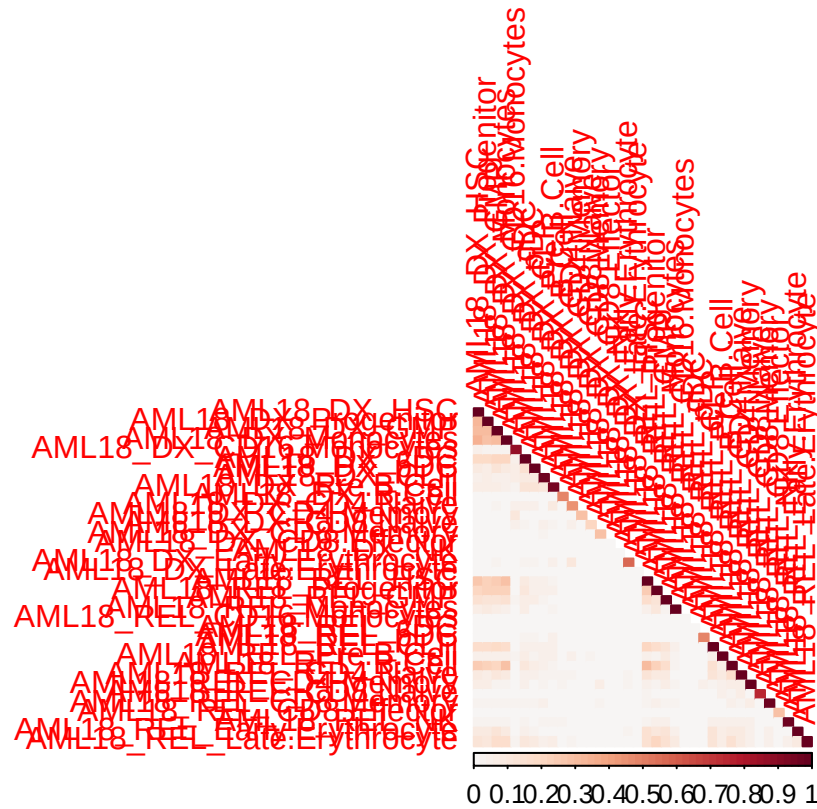
```
corrplot(mat_sorted_AML10_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```



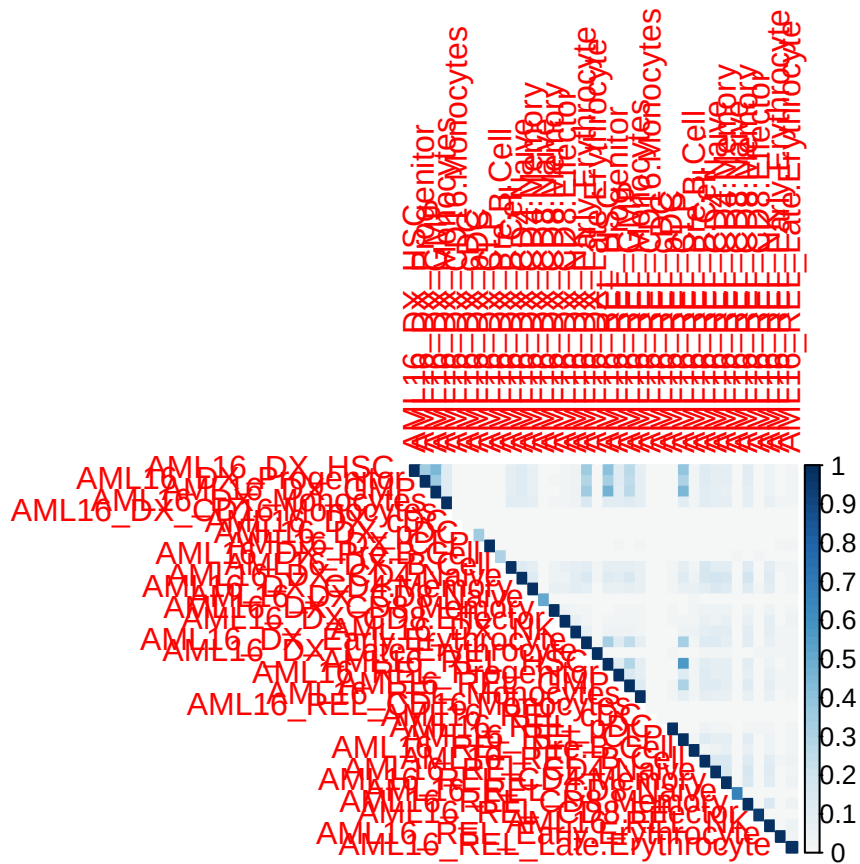
```
corrplot(mat_sorted_AML12_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```



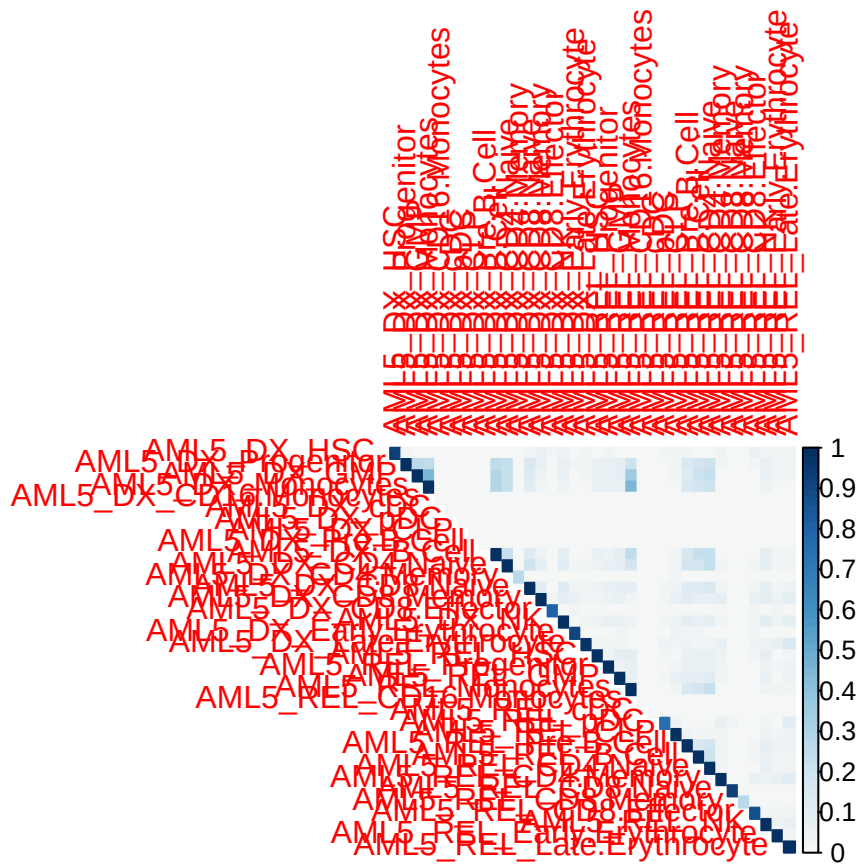
```
corrplot(mat_sorted_AML18_RNA, method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```



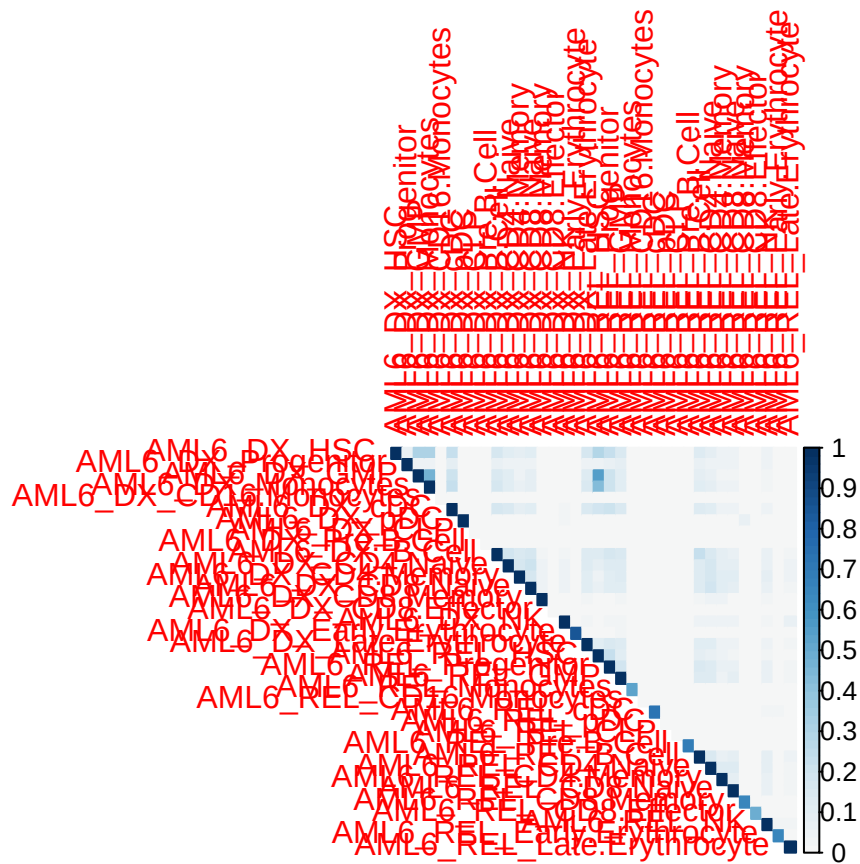
```
# ATA
corrplot(mat_sorted_AML04_ATA, method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order = )
```

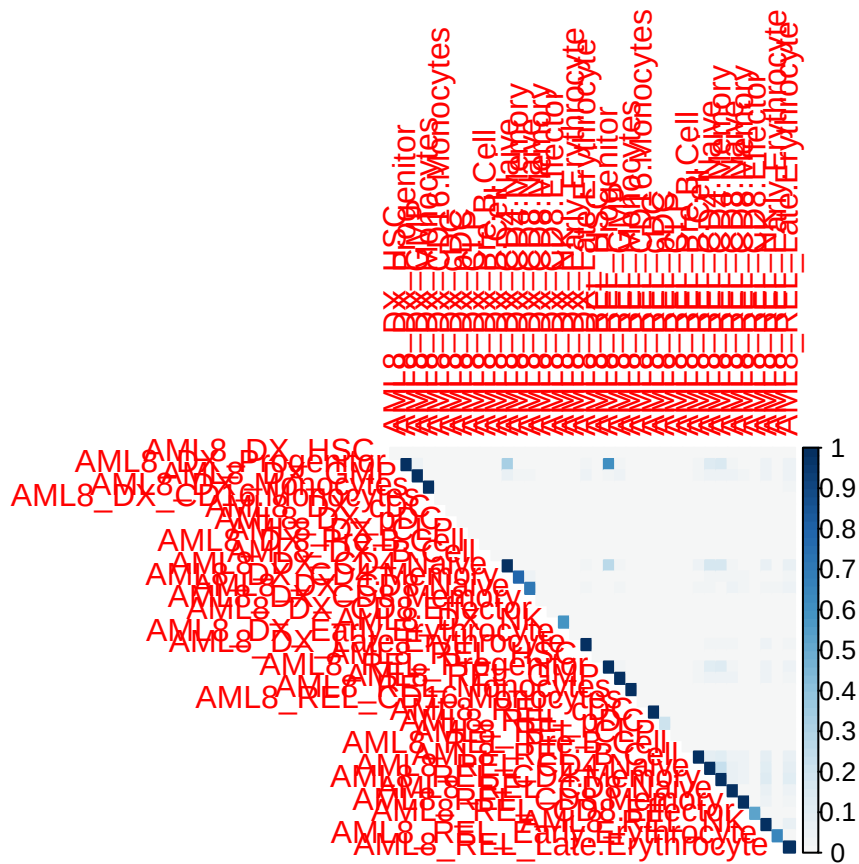
```
corrplot(mat_sorted_AML05_ATA, method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order = )
```

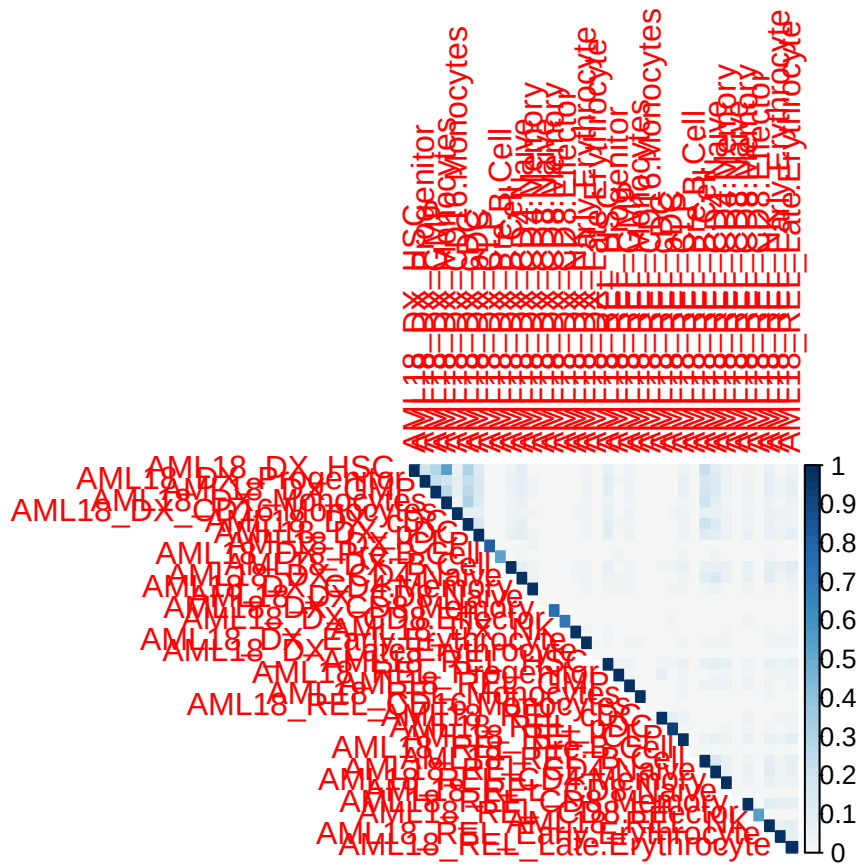
```
corrplot(mat_sorted_AML06_ATA, method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```



```
corrplot(mat_sorted_AML08_ATA, method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```

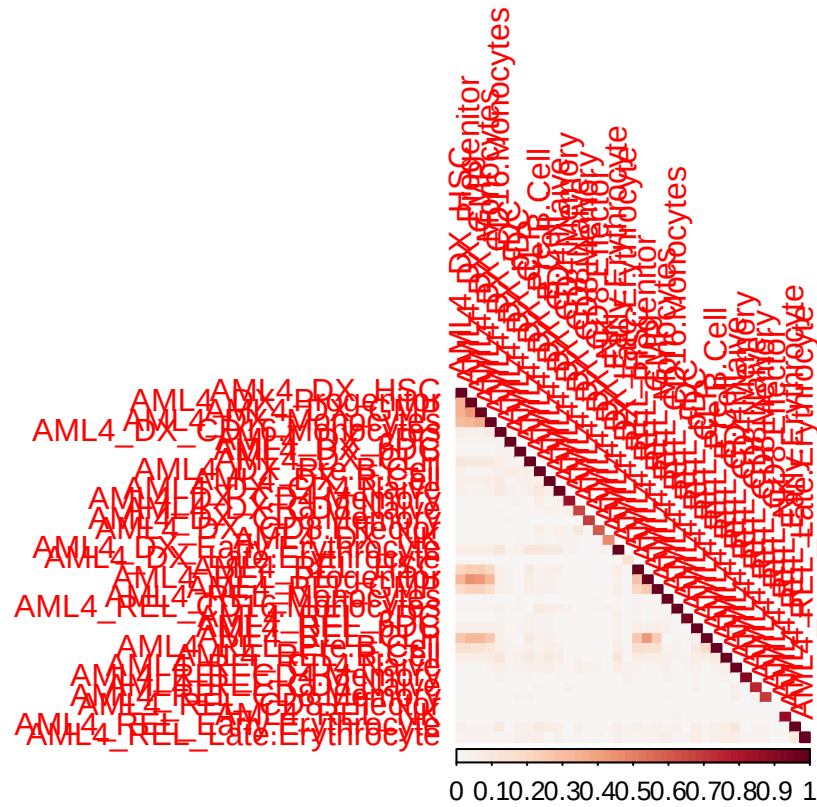


```
corrplot(mat_sorted_AML10_ATA, method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```

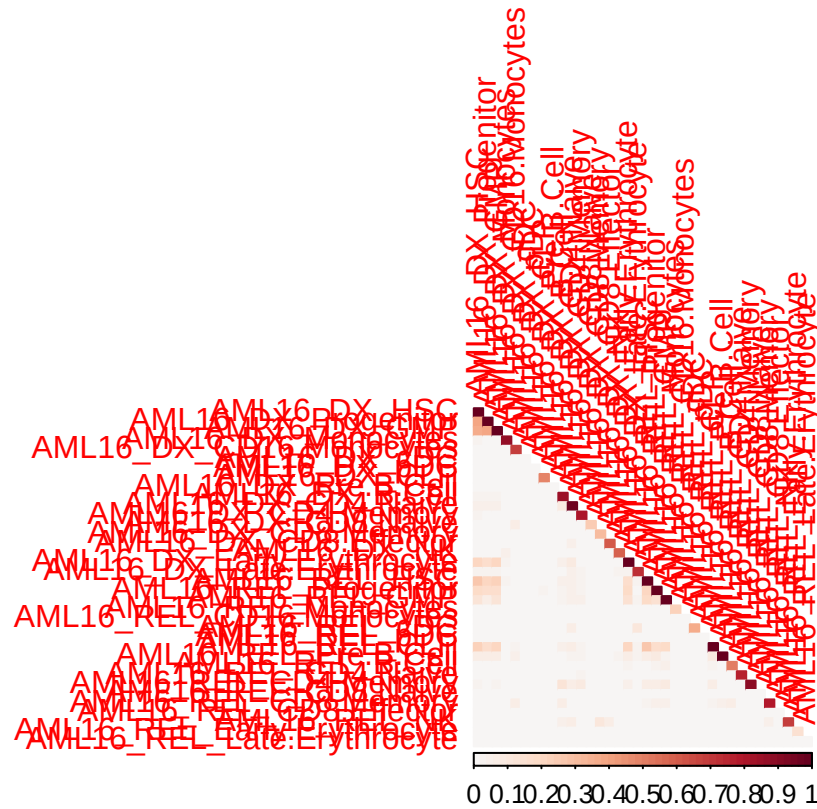



```
# RNA
```

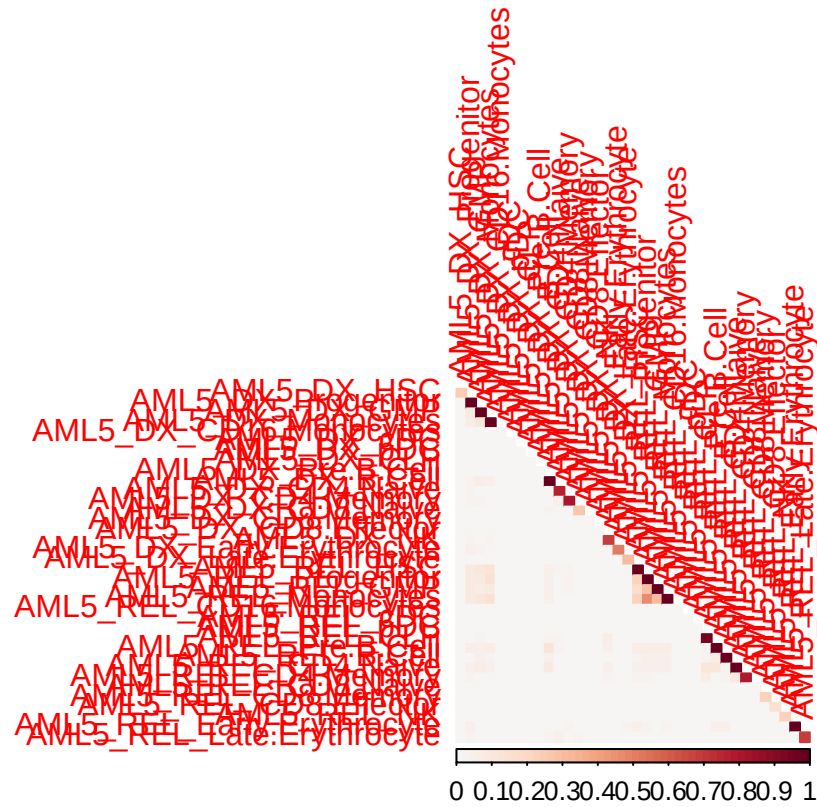
```
corrplot(mat_sorted_AML04_RNA[celltype_comm_AML04, celltype_comm_AML04], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```



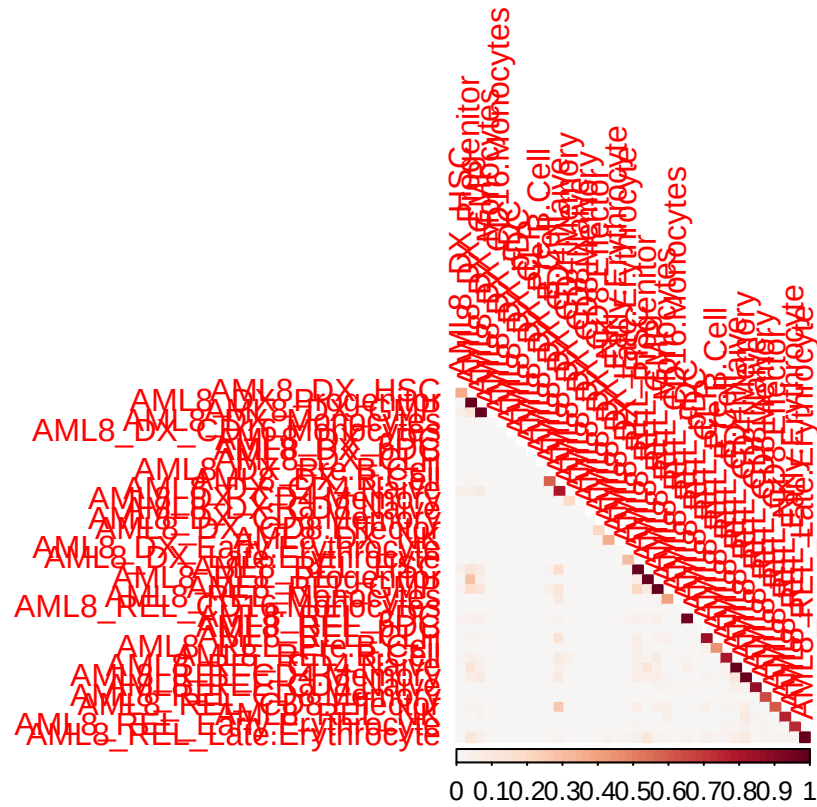
```
corrplot(mat_sorted_AML16_RNA[celltype_comm_AML16, celltype_comm_AML16], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```



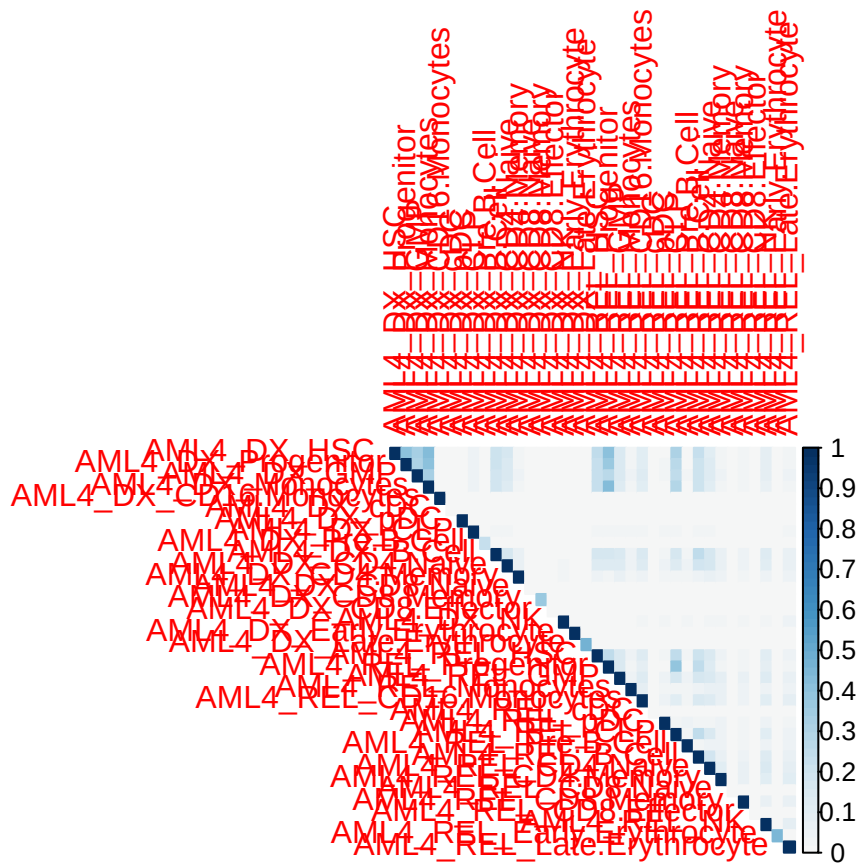
```
corrplot(mat_sorted_AML05_RNA[celltype_comm_AML05, celltype_comm_AML05], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```

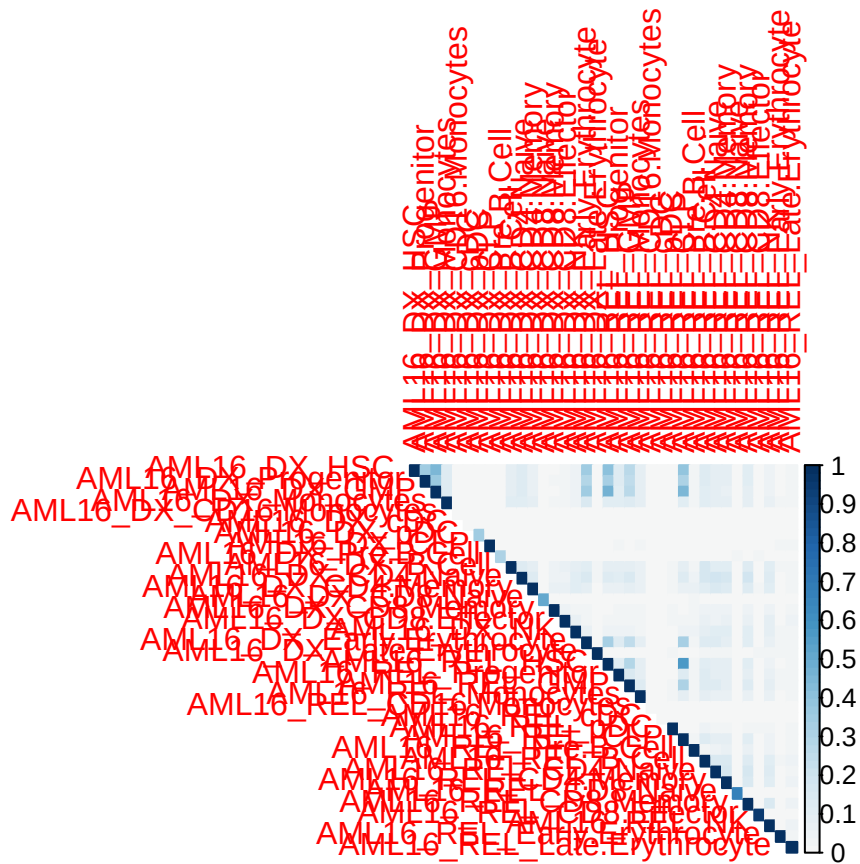
```
corrplot(mat_sorted_AML06_RNA[celltype_comm_AML06, celltype_comm_AML06], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```

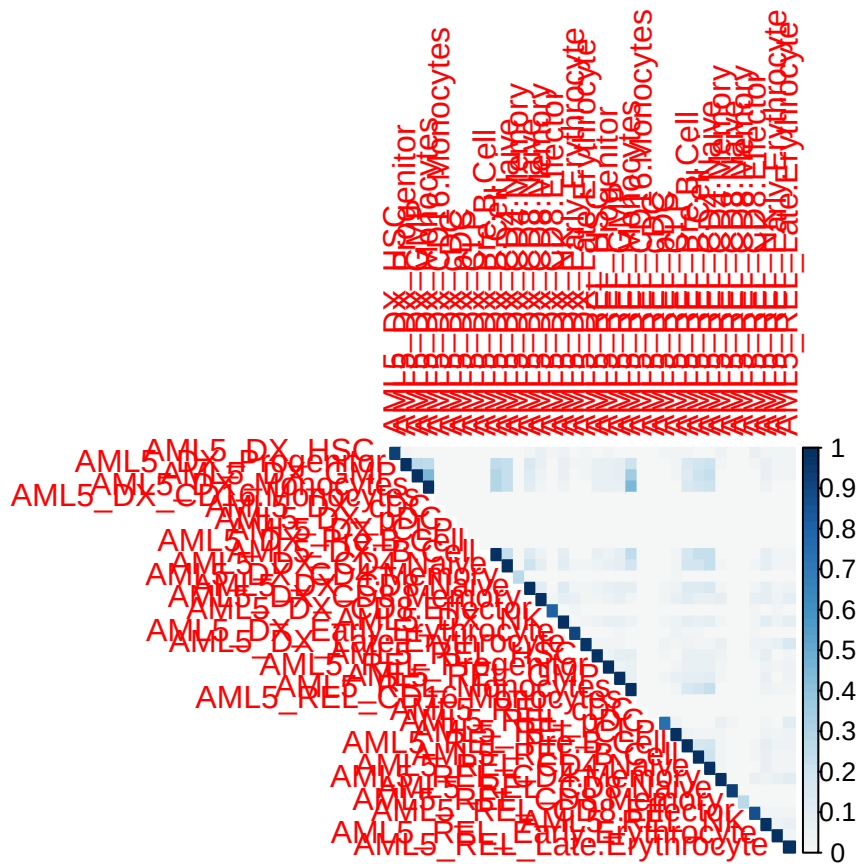
```
corrplot(mat_sorted_AML10_RNA[celltype_comm_AML10, celltype_comm_AML10], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
```

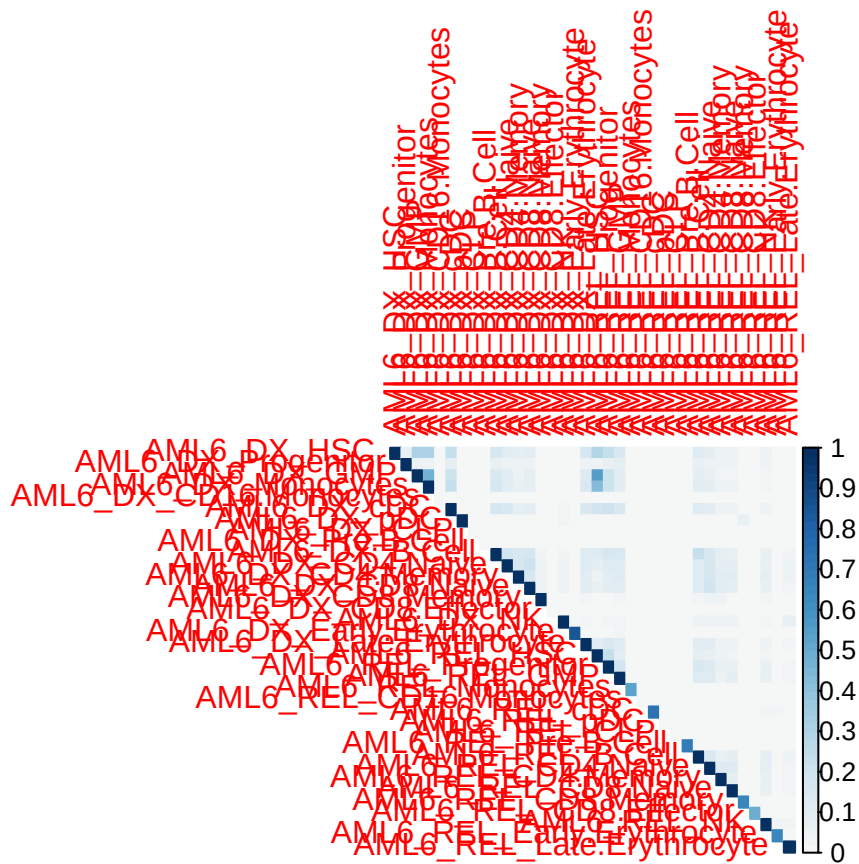
```
corrplot(mat_sorted_AML16_ATA[celltype_comm_AML16, celltype_comm_AML16], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```



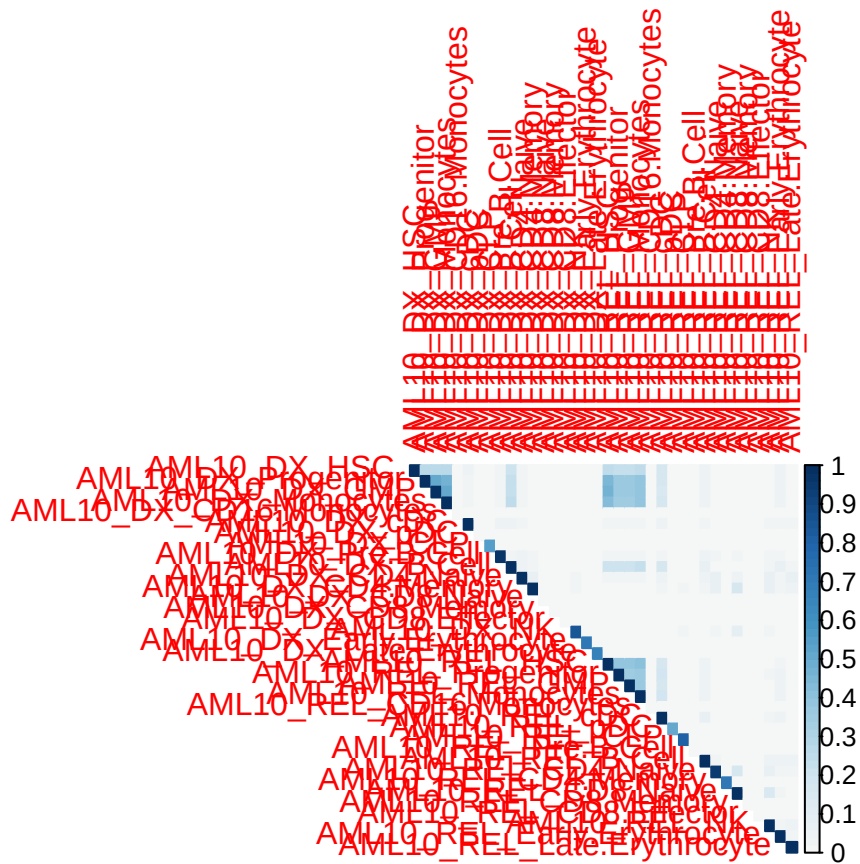
```
corrplot(mat_sorted_AML05_ATA[celltype_comm_AML05, celltype_comm_AML05], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```



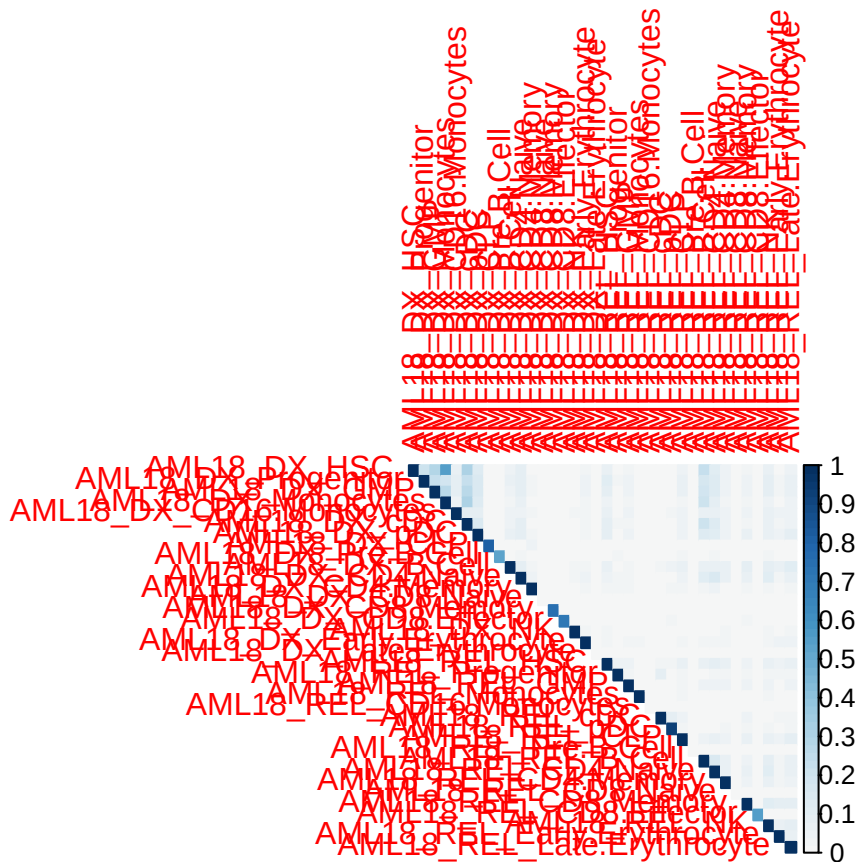
```
corrplot(mat_sorted_AML06_ATA[celltype_comm_AML06, celltype_comm_AML06], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```

```
corrplot(mat_sorted_AML08_ATA[celltype_comm_AML08, celltype_comm_AML08], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```

```
corrplot(mat_sorted_AML18_ATA[celltype_comm_AML18, celltype_comm_AML18], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
```



```
# save
pdf("../corrplot_AML04_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML04_RNA[celltype_comm_AML04, celltype_comm_AML04], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML16_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML16_RNA[celltype_comm_AML16, celltype_comm_AML16], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML05_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML05_RNA[celltype_comm_AML05, celltype_comm_AML05], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML06_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML06_RNA[celltype_comm_AML06, celltype_comm_AML06], method = 'color',
```

```

        type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML08_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML08_RNA[celltype_comm_AML08, celltype_comm_AML08], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML10_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML10_RNA[celltype_comm_AML10, celltype_comm_AML10], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML12_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML12_RNA[celltype_comm_AML12, celltype_comm_AML12], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML18_RNA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML18_RNA[celltype_comm_AML18, celltype_comm_AML18], method = 'color',
         type = 'lower', col = rev(colorRampPalette(brewer.pal(11, "RdBu"))(100)), col.lim = c(0, 1), o
dev.off()
## pdf
## 2

pdf("../corrplot_AML04_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML04_ATA[celltype_comm_AML04, celltype_comm_AML04], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML16_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML16_ATA[celltype_comm_AML16, celltype_comm_AML16], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML05_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML05_ATA[celltype_comm_AML05, celltype_comm_AML05], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

```

```

pdf("../corrplot_AML06_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML06_ATA[celltype_comm_AML06, celltype_comm_AML06], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML08_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML08_ATA[celltype_comm_AML08, celltype_comm_AML08], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML10_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML10_ATA[celltype_comm_AML10, celltype_comm_AML10], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML12_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML12_ATA[celltype_comm_AML12, celltype_comm_AML12], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

pdf("../corrplot_AML18_ATA.pdf", width = 8, height = 8)
corrplot(mat_sorted_AML18_ATA[celltype_comm_AML18, celltype_comm_AML18], method = 'color',
         type = 'upper', col = colorRampPalette(brewer.pal(11, "RdBu"))(100), col.lim = c(0, 1), order =
dev.off()
## pdf
## 2

```

7 permutation test

```

iteration_times <- 1:50
sample_size_f <- 0.50
sample_size <- 400

paired_perm_test_viz <- function(x, y, n_perm = 10000) {

  diff <- x - y
  obs <- mean(diff)

  # permutation null
  perm <- replicate(n_perm, {
    mean(diff * sample(c(-1, 1), length(diff), replace = TRUE))
  })

```

```

p <- mean(abs(perm) >= abs(obs))

# CI from null distribution
ci <- quantile(perm, c(0.025, 0.975))

# return structured object for plotting
list(
  obs = obs,
  p_value = p,
  ci = ci,
  perm_distribution = perm,
  diff_values = diff
)
}

unpaired_perm_test_viz <- function(x, y, n_perm = 10000) {

  obs <- mean(x) - mean(y)
  combined <- c(x, y)
  n_x <- length(x)

  perm <- replicate(n_perm, {
    idx <- sample(length(combined))
    mean(combined[idx[1:n_x]]) - mean(combined[idx[-(1:n_x)]])
  })

  p <- mean(abs(perm) >= abs(obs))
  ci <- quantile(perm, c(0.025, 0.975))

  list(
    obs = obs,
    p_value = p,
    ci = ci,
    perm_distribution = perm
  )
}

plot_perm_test <- function(res, title = "Permutation Test") {

  df <- data.frame(perm = res$perm_distribution)

  ggplot2::ggplot(df, ggplot2::aes(x = perm)) +
    ggplot2::geom_histogram(bins = 50, fill = "grey80", color = "black") +
    ggplot2::geom_vline(xintercept = res$obs, color = "red", size = 1) +
    ggplot2::geom_vline(xintercept = res$ci, linetype = "dashed", color = "blue") +
    ggplot2::labs(
      title = title,
      subtitle = paste0("p = ", signif(res$p_value, 3)),
      x = "Permutation mean difference",
      y = "Frequency"
    ) +
    ggplot2::theme_classic()
}

```

```

cal_sigValSimilarity <- function(mut_list, iteration_times, sampleid){
  # 1
  vals_RelLym_DxMye <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "REL_Lymphoid", sep = "_"), paste(sampleid, "D
  )

  # 2
  vals_RelMye_DxMye <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "REL_Myeloid", sep = "_"), paste(sampleid, "D
  )

  # 3
  vals_DxLym_RelMye <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "DX_Lymphoid", sep = "_"), paste(sampleid, "R
  )

  # 4
  vals_DxLym_RelLym <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "DX_Lymphoid", sep = "_"), paste(sampleid, "R
  )

  # 5
  vals_DxMye_DxLym <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "DX_Myeloid", sep = "_"), paste(sampleid, "DX
  )

  # 6
  vals_DxMye_DxLym <- sapply(
    paste0("iter_", iteration_times),
    function(x) mut_list[[sampleid]][[x]][paste(sampleid, "REL_Myeloid", sep = "_"), paste(sampleid, "R
  )

  # permutation test
  message(" 12")
  pair_12 <- unpaired_perm_test_viz(c(vals_RelLym_DxMye), c(vals_RelMye_DxMye))
  pair_12_p <- pair_12$p_value
  pair_12_c <- pair_12$obs - median(pair_12$perm_distribution)

  message(" 13")
  pair_13 <- unpaired_perm_test_viz(c(vals_RelLym_DxMye), c(vals_DxLym_RelMye))
  pair_13_p <- pair_13$p_value
  pair_13_c <- pair_13$obs - median(pair_13$perm_distribution)

  message(" 14")
  pair_14 <- unpaired_perm_test_viz(c(vals_RelLym_DxMye), c(vals_DxLym_RelLym))
  pair_14_p <- pair_14$p_value

```



```

pair_14_c <- pair_14$obs - median(pair_14$perm_distribution)

message(" 15")
pair_15 <- unpaired_perm_test_viz(c(vals_RelLym_DxMye), c(vals_DxMye_DxLym))
pair_15_p <- pair_15$p_value
pair_15_c <- pair_15$obs - median(pair_15$perm_distribution)

message(" 16")
pair_16 <- unpaired_perm_test_viz(c(vals_RelLym_DxMye), c(vals_DxMye_DxLym))
pair_16_p <- pair_16$p_value
pair_16_c <- pair_16$obs - median(pair_16$perm_distribution)

result <- list(
  pair_12_p = pair_12_p,
  pair_12_c = pair_12_c,
  pair_13_p = pair_13_p,
  pair_13_c = pair_13_c,
  pair_14_p = pair_14_p,
  pair_14_c = pair_14_c,
  pair_15_p = pair_15_p,
  pair_15_c = pair_15_c,
  pair_16_p = pair_16_p,
  pair_16_c = pair_16_c
)

return(result)
}

selected_samples_id <- c("AML2", "AML4", "AML5", "AML6", "AML8", "AML10", "AML11", "AML12",
  "AML13", "AML15", "AML16", "AML17", "AML18", "AML19", "AML21",
  "AML22", "AML24", "AML25", "AML26", "AML27")

# RNA
data_scRNA_sigSimilarity <- as.data.frame(matrix(data = 0, nrow = length(selected_samples_id), ncol = 16))
rownames(data_scRNA_sigSimilarity) <- selected_samples_id
colnames(data_scRNA_sigSimilarity) <- paste(c("pair_12_p", "pair_12_c", "pair_13_p", "pair_13_c",
  "pair_14_p", "pair_14_c", "pair_15_p", "pair_15_c",
  "pair_16_p", "pair_16_c"), "RNA", sep = "_")

# ATA
data_scATA_sigSimilarity <- as.data.frame(matrix(data = 0, nrow = length(selected_samples_id), ncol = 16))
rownames(data_scATA_sigSimilarity) <- selected_samples_id
colnames(data_scATA_sigSimilarity) <- paste(c("pair_12_p", "pair_12_c", "pair_13_p", "pair_13_c",
  "pair_14_p", "pair_14_c", "pair_15_p", "pair_15_c",
  "pair_16_p", "pair_16_c"), "ATA", sep = "_")

for (sampleid in selected_samples_id){
  message(paste0(sampleid, "-scRNA"))
  r_scRNA <- cal_sigValSimilarity(mut_cellTypeAnn_similarity_list_RNA_main, iteration_times, sampleid)
  message(paste0(sampleid, "-scATA"))
  r_scATA <- cal_sigValSimilarity(mut_cellTypeAnn_similarity_list_ATA_main, iteration_times, sampleid)
  data_scRNA_sigSimilarity[sampleid, "pair_12_p_RNA"] <- r_scRNA$pair_12_p
  data_scRNA_sigSimilarity[sampleid, "pair_12_c_RNA"] <- r_scRNA$pair_12_c
  data_scRNA_sigSimilarity[sampleid, "pair_13_p_RNA"] <- r_scRNA$pair_13_p

```

```

data_scRNA_sigSimilarity[sampleid, "pair_13_c_RNA"] <- r_scRNA$pair_13_c
data_scRNA_sigSimilarity[sampleid, "pair_14_p_RNA"] <- r_scRNA$pair_14_p
data_scRNA_sigSimilarity[sampleid, "pair_14_c_RNA"] <- r_scRNA$pair_14_c
data_scRNA_sigSimilarity[sampleid, "pair_15_p_RNA"] <- r_scRNA$pair_15_p
data_scRNA_sigSimilarity[sampleid, "pair_15_c_RNA"] <- r_scRNA$pair_15_c
data_scRNA_sigSimilarity[sampleid, "pair_16_p_RNA"] <- r_scRNA$pair_16_p
data_scRNA_sigSimilarity[sampleid, "pair_16_c_RNA"] <- r_scRNA$pair_16_c

data_scATA_sigSimilarity[sampleid, "pair_12_p_ATA"] <- r_scATA$pair_12_p
data_scATA_sigSimilarity[sampleid, "pair_12_c_ATA"] <- r_scATA$pair_12_c
data_scATA_sigSimilarity[sampleid, "pair_13_p_ATA"] <- r_scATA$pair_13_p
data_scATA_sigSimilarity[sampleid, "pair_13_c_ATA"] <- r_scATA$pair_13_c
data_scATA_sigSimilarity[sampleid, "pair_14_p_ATA"] <- r_scATA$pair_14_p
data_scATA_sigSimilarity[sampleid, "pair_14_c_ATA"] <- r_scATA$pair_14_c
data_scATA_sigSimilarity[sampleid, "pair_15_p_ATA"] <- r_scATA$pair_15_p
data_scATA_sigSimilarity[sampleid, "pair_15_c_ATA"] <- r_scATA$pair_15_c
data_scATA_sigSimilarity[sampleid, "pair_16_p_ATA"] <- r_scATA$pair_16_p
data_scATA_sigSimilarity[sampleid, "pair_16_c_ATA"] <- r_scATA$pair_16_c
}

## AML2-scrNA
## 12
## 13
## 14
## 15
## 16
## AML2-scATA
## 12
## 13
## 14
## 15
## 16
## AML4-scrNA
## 12
## 13
## 14
## 15
## 16
## AML4-scATA
## 12
## 13
## 14
## 15
## 16
## AML5-scrNA
## 12
## 13
## 14
## 15
## 16
## AML5-scATA
## 12
## 13

```

```
## 14
## 15
## 16
## AML6-scrNA
## 12
## 13
## 14
## 15
## 16
## AML6-scATA
## 12
## 13
## 14
## 15
## 16
## AML8-scrNA
## 12
## 13
## 14
## 15
## 16
## AML8-scATA
## 12
## 13
## 14
## 15
## 16
## AML10-scrNA
## 12
## 13
## 14
## 15
## 16
## AML10-scATA
## 12
## 13
## 14
## 15
## 16
## AML11-scrNA
## 12
## 13
## 14
## 15
## 16
## AML11-scATA
## 12
## 13
## 14
## 15
## 16
## AML12-scrNA
## 12
```

```
## 13
## 14
## 15
## 16
## AML12-scATA
## 12
## 13
## 14
## 15
## 16
## AML13-scRNA
## 12
## 13
## 14
## 15
## 16
## AML13-scATA
## 12
## 13
## 14
## 15
## 16
## AML15-scRNA
## 12
## 13
## 14
## 15
## 16
## AML15-scATA
## 12
## 13
## 14
## 15
## 16
## AML16-scRNA
## 12
## 13
## 14
## 15
## 16
## AML16-scATA
## 12
## 13
## 14
## 15
## 16
## AML17-scRNA
## 12
## 13
## 14
## 15
## 16
## AML17-scATA
```

```

## 12
## 13
## 14
## 15
## 16
## AML18-scRNA
## 12
## 13
## 14
## 15
## 16
## AML18-scATA
## 12
## 13
## 14
## 15
## 16
## AML19-scRNA
## 12
## 13
## 14
## 15
## 16
## AML19-scATA
## 12
## 13
## 14
## 15
## 16
## AML21-scRNA
## 12
## 13
## 14
## 15
## 16
## AML21-scATA
## 12
## 13
## 14
## 15
## 16
## AML22-scRNA
## 12
## 13
## 14
## 15
## 16
## AML22-scATA
## 12
## 13
## 14
## 15
## 16

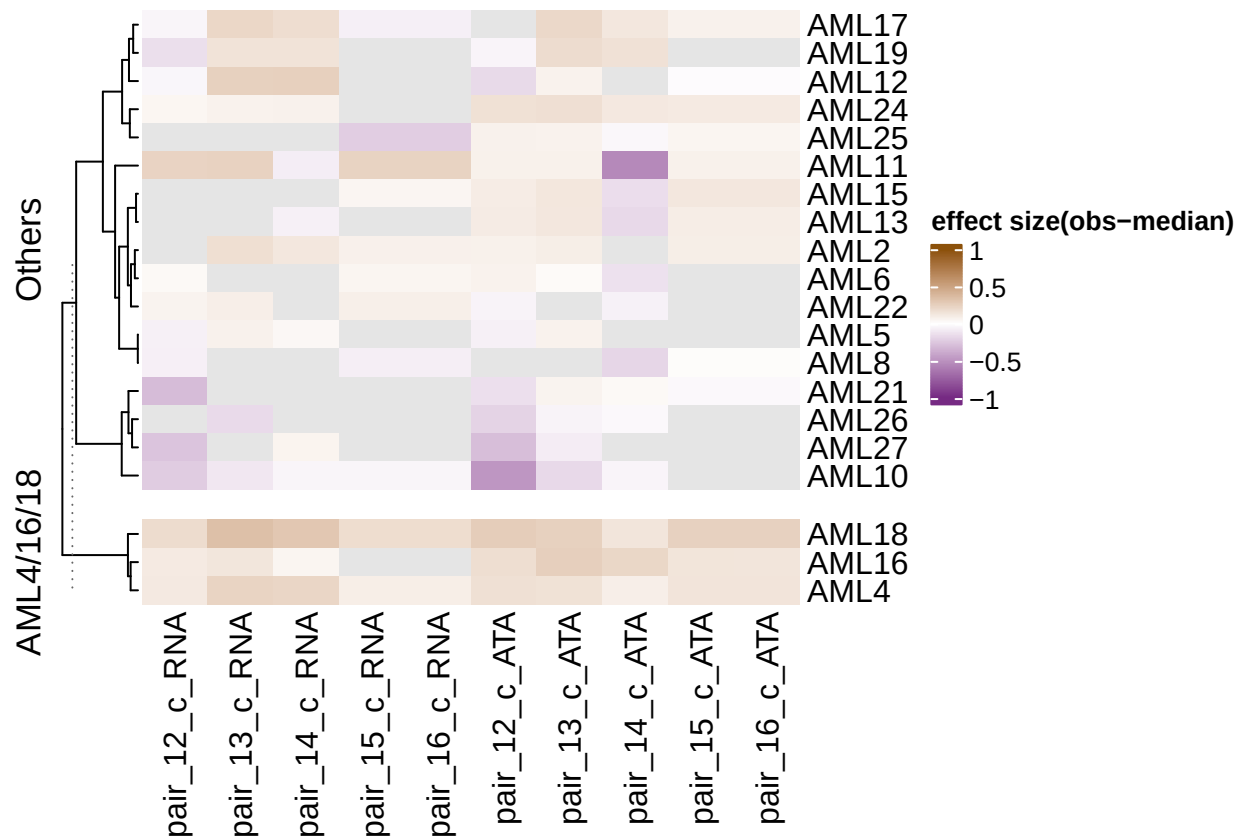
```

```

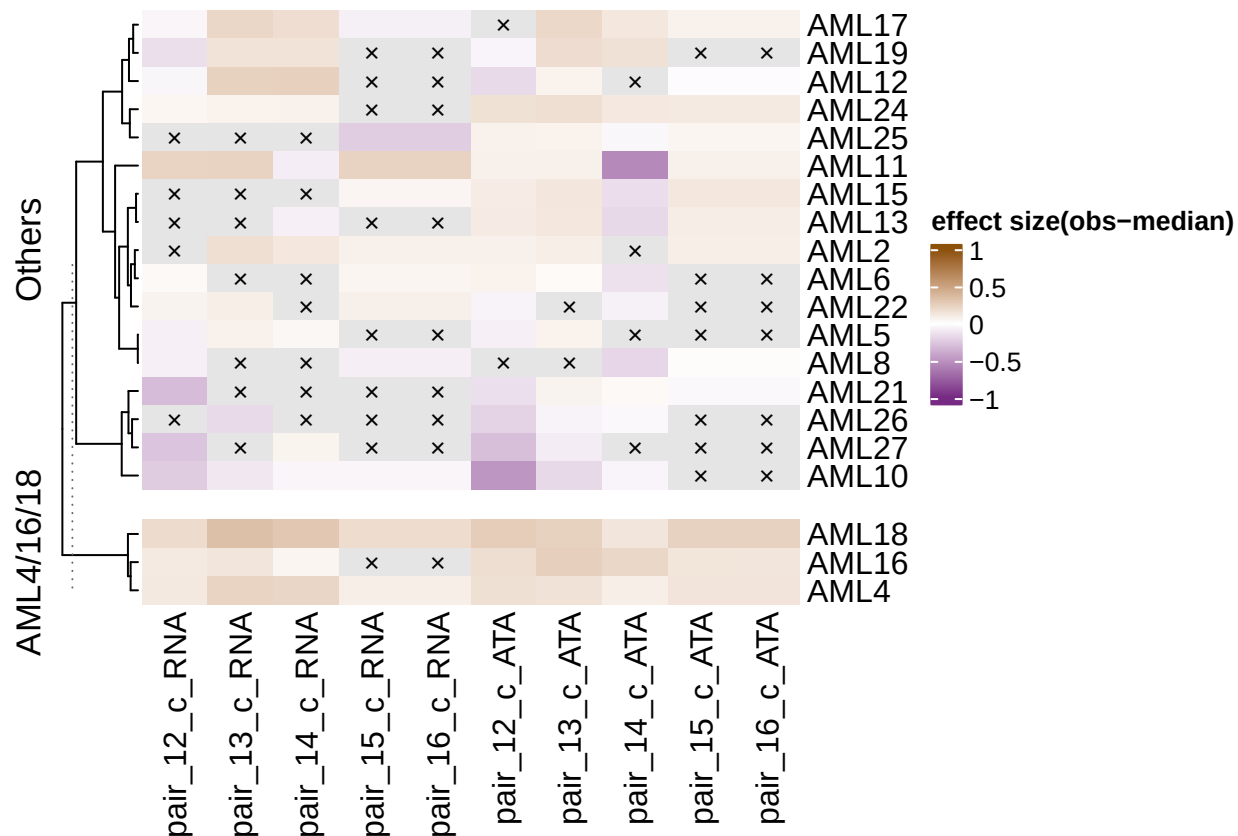
## AML24-scRNA
## 12
## 13
## 14
## 15
## 16
## AML24-scATA
## 12
## 13
## 14
## 15
## 16
## AML25-scRNA
## 12
## 13
## 14
## 15
## 16
## AML25-scATA
## 12
## 13
## 14
## 15
## 16
## AML26-scRNA
## 12
## 13
## 14
## 15
## 16
## AML26-scATA
## 12
## 13
## 14
## 15
## 16
## AML27-scRNA
## 12
## 13
## 14
## 15
## 16
## AML27-scATA
## 12
## 13
## 14
## 15
## 16

df_RNA <- data_scRNA_sigSimilarity
pairs_RNA <- unique(sub("_p_RNA", "", grep("_p_RNA$", colnames(df_RNA), value = TRUE)))
for (p in pairs_RNA) {
  p_col <- paste0(p, "_p_RNA")

```

```
Heatmap(
  mat,
  name = "effect size(obs-median)",
  row_split = group,
  cluster_rows = TRUE,
  clustering_method_rows = "ward.D",
  cluster_columns = FALSE,
  col = col_fun,
  show_row_names = TRUE,
  row_gap = unit(4, "mm"),
  na_col = "grey90",
  cell_fun = function(j, i, x, y, w, h, fill) {
    if (is.na(mat[i, j])) {
      grid::grid.text(
        "x", x = x, y = y,
        gp = grid::gpar(fontsize = 10, col = "black")
      )
    }
  }
)
## Warning: The input is a data frame-like object, convert it to a matrix.
```

```
ht <- Heatmap(
  mat,
  name = "effect size(obs-median)",
  row_split = group,
  cluster_rows = TRUE,
  clustering_method_rows = "ward.D",
  cluster_columns = FALSE,
  col = col_fun,
  show_row_names = TRUE,
  row_gap = unit(4, "mm"),
  na_col = "grey90"
)
## Warning: The input is a data frame-like object, convert it to a matrix.

pdf("heatmap_effect_size.pdf", width = 5.8, height = 6)

draw(ht)

dev.off()
## pdf
## 2

sample_selected <- "AML18"
```

```

# 1
vals_RelLym_DxMye_RNA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_RNA_main[[sample_selected]][[x]][paste(sample_selected, "
)

vals_RelLym_DxMye_ATA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_ATA_main[[sample_selected]][[x]][paste(sample_selected, "
)

# 2
vals_RelLym_DxLym_RNA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_RNA_main[[sample_selected]][[x]][paste(sample_selected, "
)

vals_RelLym_DxLym_ATA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_ATA_main[[sample_selected]][[x]][paste(sample_selected, "
)

# 3
vals_DxMye_DxLym_RNA<- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_RNA_main[[sample_selected]][[x]][paste(sample_selected, "
)

vals_DxMye_DxLym_ATA<- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_ATA_main[[sample_selected]][[x]][paste(sample_selected, "
)

# 4
vals_RelMye_RelLym_RNA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_RNA_main[[sample_selected]][[x]][paste(sample_selected,
)
vals_RelMye_RelLym_ATA <- sapply(
  paste0("iter_", iteration_times),
  function(x) mut_cellTypeAnn_similarity_list_ATA_main[[sample_selected]][[x]][paste(sample_selected,
)

set.seed(2025)
res_RNA12 <- unpaired_perm_test_viz(vals_RelLym_DxMye_RNA, vals_RelLym_DxLym_RNA)
res_RNA13 <- unpaired_perm_test_viz(vals_RelLym_DxMye_RNA, vals_DxMye_DxLym_RNA)
res_RNA14 <- unpaired_perm_test_viz(vals_RelLym_DxMye_RNA, vals_RelMye_RelLym_RNA)

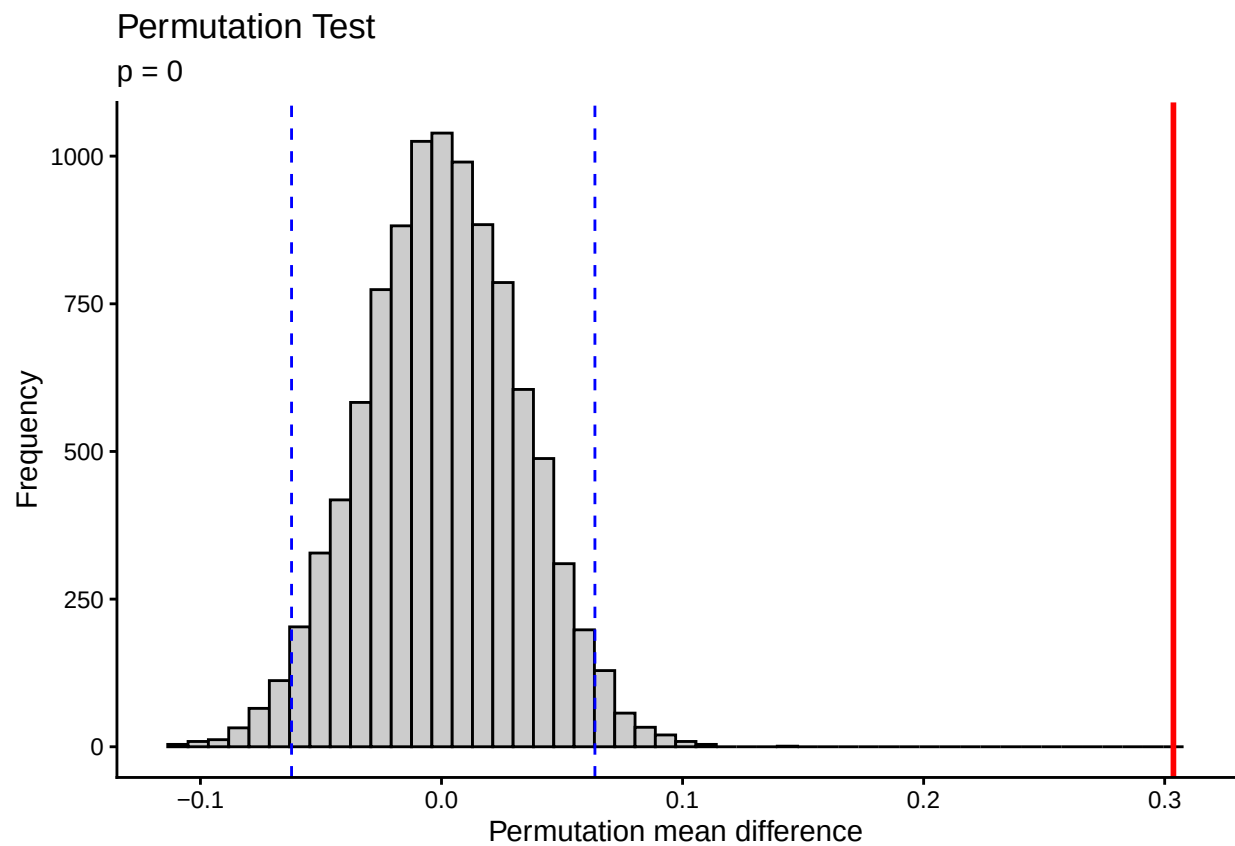
res_ATA12 <- unpaired_perm_test_viz(vals_RelLym_DxMye_ATA, vals_RelLym_DxLym_ATA)
res_ATA13 <- unpaired_perm_test_viz(vals_RelLym_DxMye_ATA, vals_DxMye_DxLym_ATA)
res_ATA14 <- unpaired_perm_test_viz(vals_RelLym_DxMye_ATA, vals_RelMye_RelLym_ATA)

p_res_RNA12 <- plot_perm_test(res_RNA12)
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.

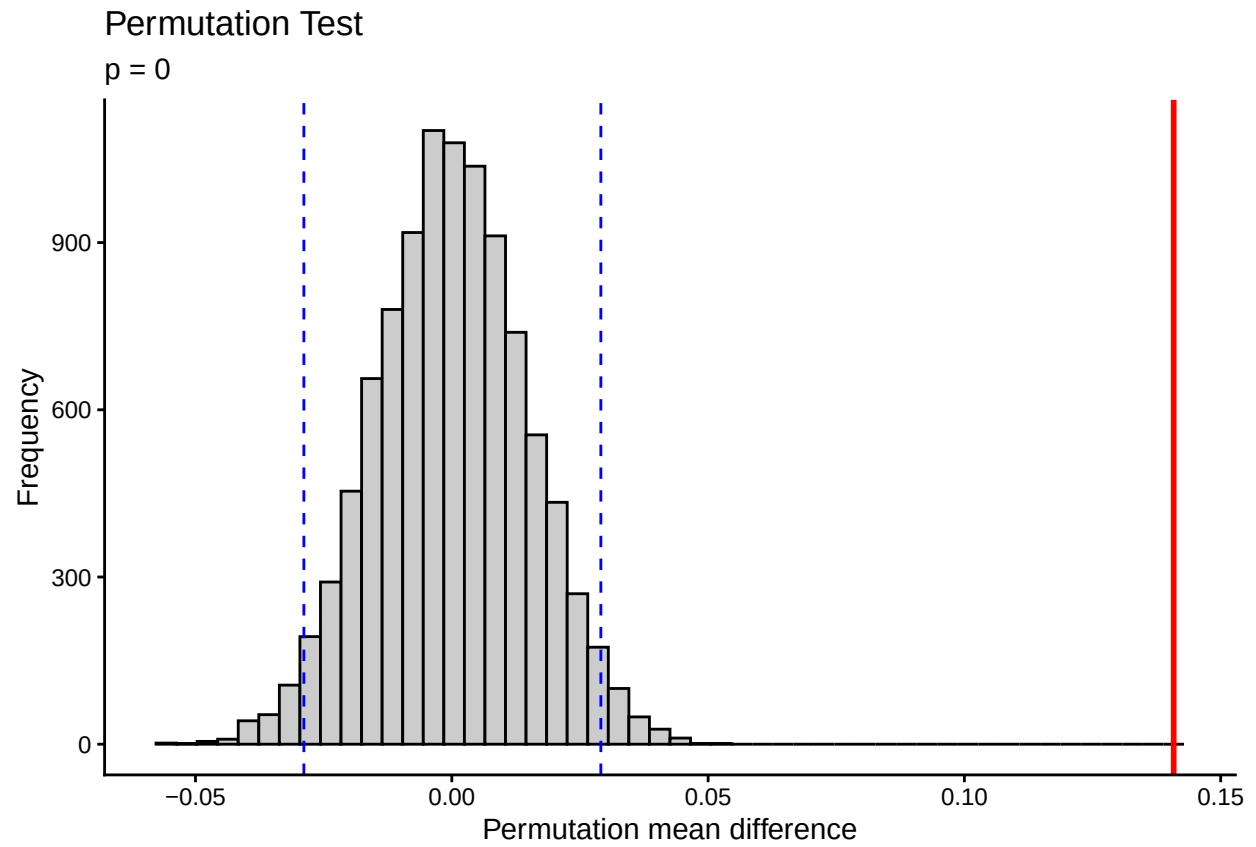
```

```
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
p_res_ATA12 <- plot_perm_test(res_ATA12)
p_res_RNA13 <- plot_perm_test(res_RNA13)
p_res_ATA13 <- plot_perm_test(res_ATA13)
p_res_RNA14 <- plot_perm_test(res_RNA14)
p_res_ATA14 <- plot_perm_test(res_ATA14)

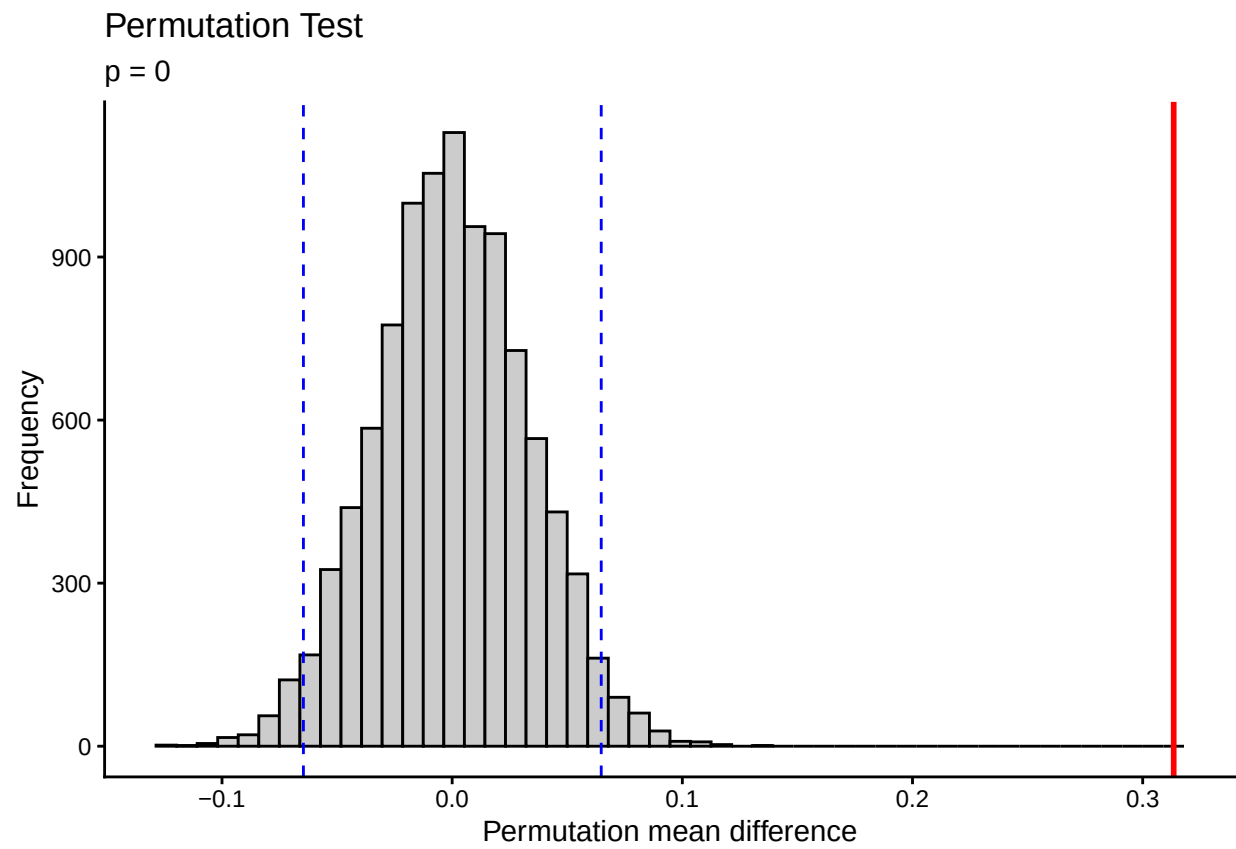
p_res_RNA12
```



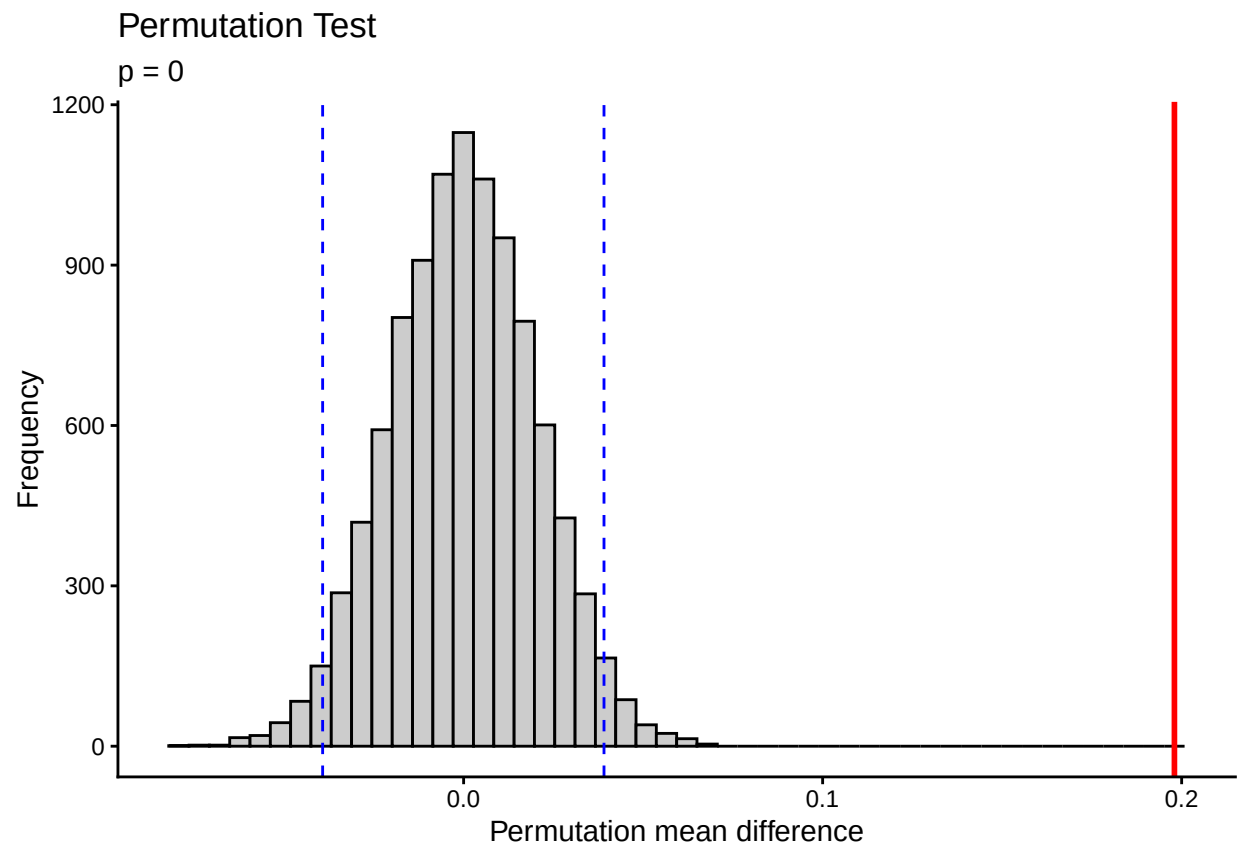
```
p_res_ATA12
```



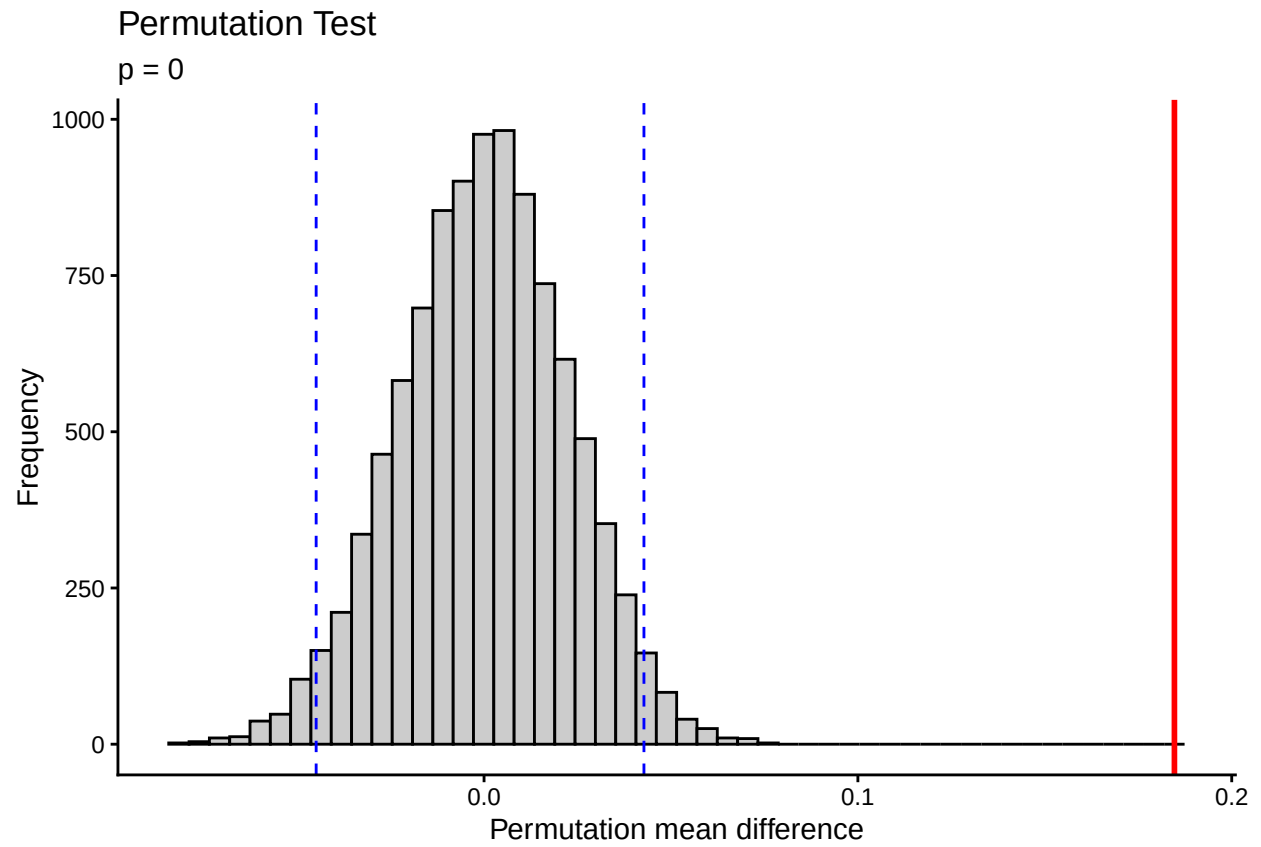
p_res_RNA13



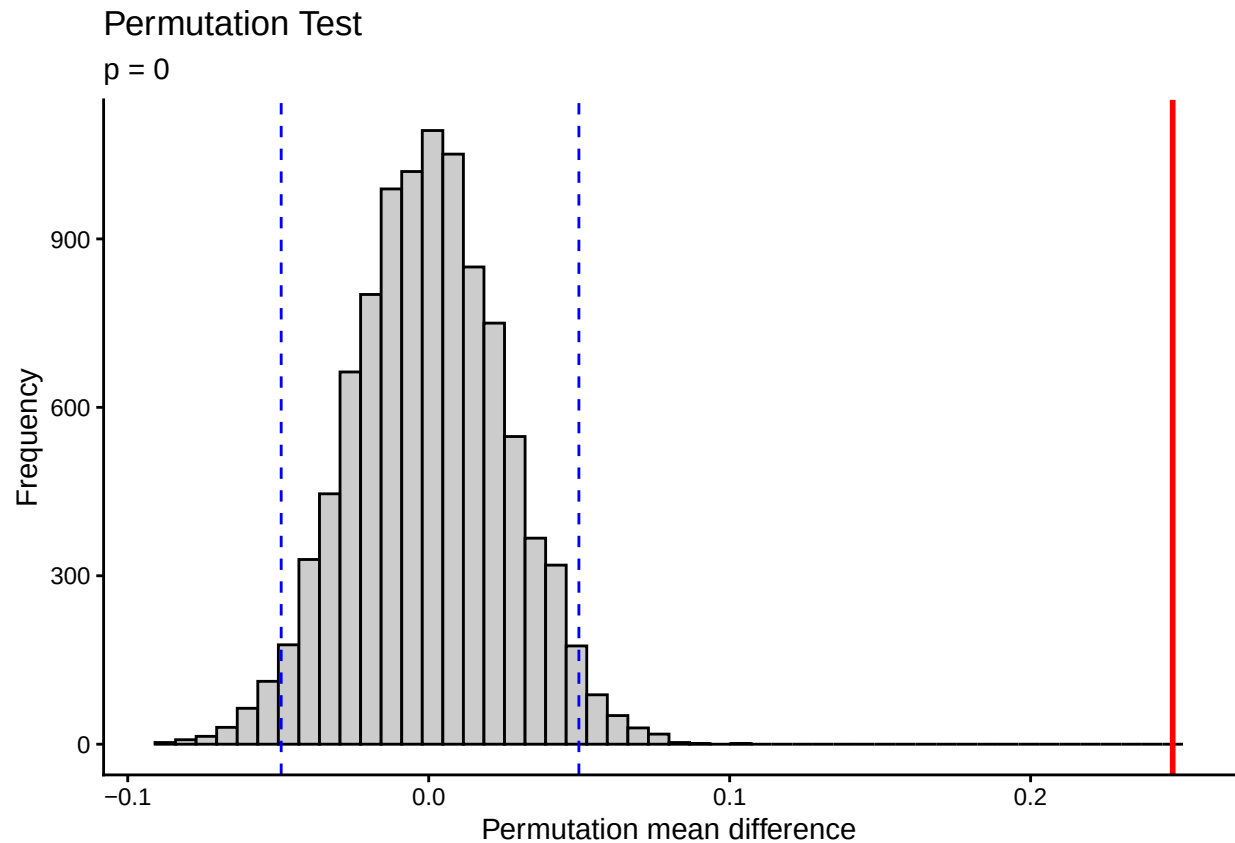
p_res_ATA13



p_res_RNA14



p_res_ATA14



```
data_value_similarity_RNA <- list(DXMy_RELly = vals_RelLym_DxMye_RNA,
                                DXLy_RELly = vals_RelLym_DxLym_RNA,
                                DXMy_DXLy = vals_DxMye_DxLym_RNA,
                                RELMy_RELly = vals_RelMye_RelLym_RNA)
```

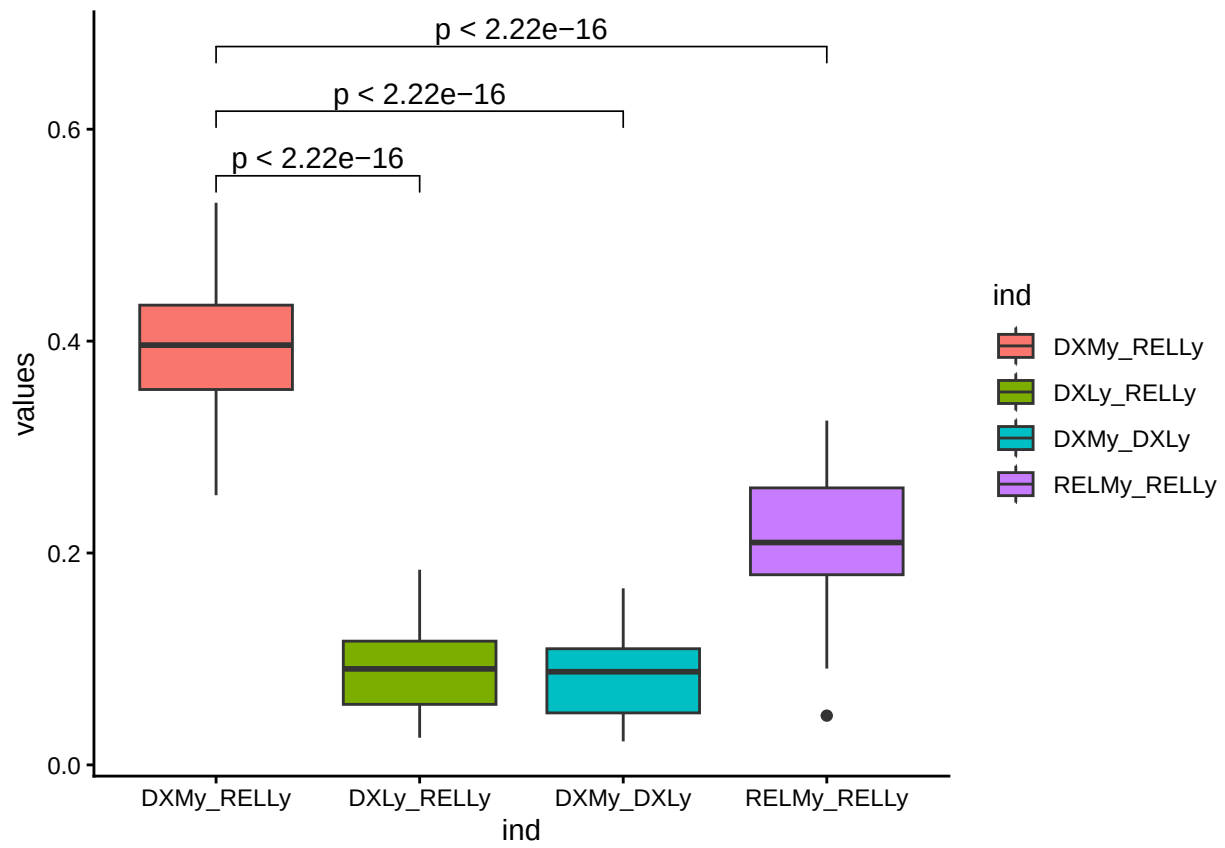
```
data_value_similarity_ATA <- list(DXMy_RELly = vals_RelLym_DxMye_ATA,
                                  DXLy_RELly = vals_RelLym_DxLym_ATA,
                                  DXMy_DXLy = vals_DxMye_DxLym_ATA,
                                  RELMy_RELly = vals_RelMye_RelLym_ATA)
```

```
comparisons <- list(
  c("DXMy_RELly", "DXLy_RELly"),
  c("DXMy_RELly", "DXMy_DXLy"),
  c("DXMy_RELly", "RELMy_RELly")
)
```

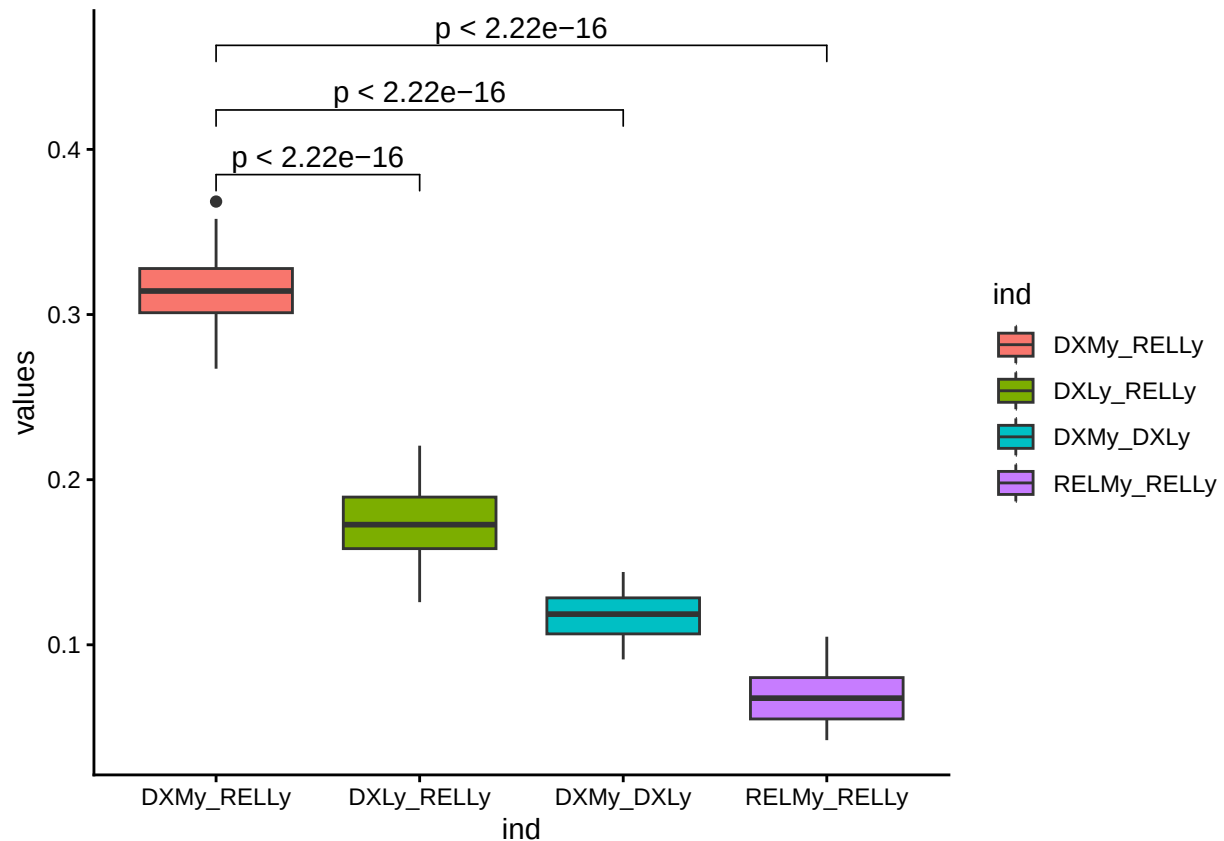
```
p_boxPlotdata_value_similarity_RNA <- ggplot(stack(data_value_similarity_RNA), aes(ind, values, fill = )) +
  geom_boxplot() +
  stat_compare_means(comparisons = comparisons, method = "wilcox.test") +
  theme_classic()
```

```
p_boxPlotdata_value_similarity_ATA <- ggplot(stack(data_value_similarity_ATA), aes(ind, values, fill = )) +
  geom_boxplot() +
  stat_compare_means(comparisons = comparisons, method = "wilcox.test") +
  theme_classic()
```


p_boxPlotdata_value_similarity_RNA



p_boxPlotdata_value_similarity_ATA



```
#ggsave("../p_boxPlotdata_value_similarity_RNA.pdf", p_boxPlotdata_value_similarity_RNA, width = 3.2, height = 2.2)
#ggsave("../p_boxPlotdata_value_similarity_ATA.pdf", p_boxPlotdata_value_similarity_ATA, width = 3.2, height = 2.2)
```

8 mutations shared by different lineage

```
cellxmut_to_01 <- function(cellxmut,
                           mapping = c(NoCall = 0L, WT = 0L, Mut = 1L),
                           default_value = NA_integer_,
                           return_matrix = TRUE) {
  if (!(is.matrix(cellxmut) || is.data.frame(cellxmut))) {
    stop("cellxmut must be a matrix or data.frame.")
  }

  rn <- rownames(cellxmut)
  cn <- colnames(cellxmut)
  if (is.null(rn) || any(rn == "")) stop("cellxmut must have non-empty rownames (cell names).")
  if (is.null(cn) || any(cn == "")) stop("cellxmut must have non-empty colnames (mutation sites).")

  # Work in character space, trim whitespace
  x <- as.matrix(cellxmut)
  x <- apply(x, c(1, 2), function(v) trimws(as.character(v)))
}
```

```

# If already numeric-ish (e.g., "0"/"1"), keep them
suppressWarnings(num <- matrix(as.integer(x), nrow = nrow(x), ncol = ncol(x))
is_num <- !is.na(num) & x %in% c("0", "1")
out <- matrix(NA_integer_, nrow = nrow(x), ncol = ncol(x), dimnames = list(rn, cn))
out[is_num] <- num[is_num]

# Map remaining via mapping
map_vec <- as.integer(mapping); names(map_vec) <- names(mapping)
idx <- is.na(out)
if (any(idx)) {
  mapped <- map_vec[x[idx]]
  mapped[is.na(mapped)] <- default_value
  out[idx] <- as.integer(mapped)
}

if (return_matrix) return(out)
as.data.frame(out, check.names = FALSE)
}

celltype_matrix <- function(mat, meta, method = "sum") {

  # 1. clean cell names
  cells_format_mat <- sub("_L[0-9]+_", "_", rownames(mat))

  # 2. mapping
  meta_map <- setNames(meta$maincelltype, meta$CellTag)

  # 3. cells
  keep_idx <- cells_format_mat %in% meta$CellTag

  mat <- mat[keep_idx, , drop = FALSE]
  cells_use <- cells_format_mat[keep_idx]

  celltype <- meta_map[cells_use]

  # 4. attach celltype
  df <- as.data.frame(mat)
  df$maincelltype <- celltype

  # 5. aggregation
  if (method == "sum") {
    agg <- aggregate(. ~ maincelltype, data = df, FUN = sum)
  } else if (method == "mean") {
    agg <- aggregate(. ~ maincelltype, data = df, FUN = mean)
  } else if (method == "binary") {
    agg <- aggregate(. ~ maincelltype, data = df, FUN = function(x) as.numeric(any(x > 0)))
  } else {
    stop("method must be 'sum', 'mean' or 'binary'")
  }

  rownames(agg) <- agg$maincelltype
  agg$maincelltype <- NULL
}

```

```

    return(as.matrix(agg))
}

# scRNA
celltypes <- c("HSPC", "Myeloid", "Lymphoid", "Erythroid")
samples_scRNA_csm <- intersect(names(AML_scRNA_CSM_list), unique(AML_scRNA_meta$Lambo_et_al_ID))

mutation_ct_scRNA <- vector("list", length(samples_scRNA_csm))
names(mutation_ct_scRNA) <- samples_scRNA_csm
for (sample in samples_scRNA_csm){
  message(sample)
  csm_01 <- cellxmut_to_01(AML_scRNA_CSM_list[[sample]])
  csm_01_ct <- celltype_matrix(mat = csm_01, meta = AML_scRNA_meta[AML_scRNA_meta$Lambo_et_al_ID == sample,])
  csm_01_ct <- csm_01_ct[celltypes,]
  csm_01_ct_bin <- ifelse(csm_01_ct >= 3, 1, 0)
  df <- as.data.frame(t(csm_01_ct_bin))
  df <- df %>% rownames_to_column("mutation")
  df[celltypes] <- lapply(df[celltypes], function(x) as.logical(as.integer(x)))
  mutation_ct_scRNA[[sample]] <- df
}

## AML1
## AML2
## AML3
## AML4
## AML5
## AML6
## AML7
## AML8
## AML9
## AML10
## AML11
## AML12
## AML13
## AML14
## AML15
## AML16
## AML17
## AML18
## AML19
## AML20
## AML21
## AML22
## AML23
## AML24
## AML25
## AML26
## AML27
## AML28

# scATA
celltypes <- c("HSPC", "Myeloid", "Lymphoid", "Erythroid")
samples_scATA_csm <- intersect(names(AML_scATA_CSM_list), unique(AML_scATA_meta$SampleID))
mutation_ct_scATA <- vector("list", length(samples_scATA_csm))

```

```

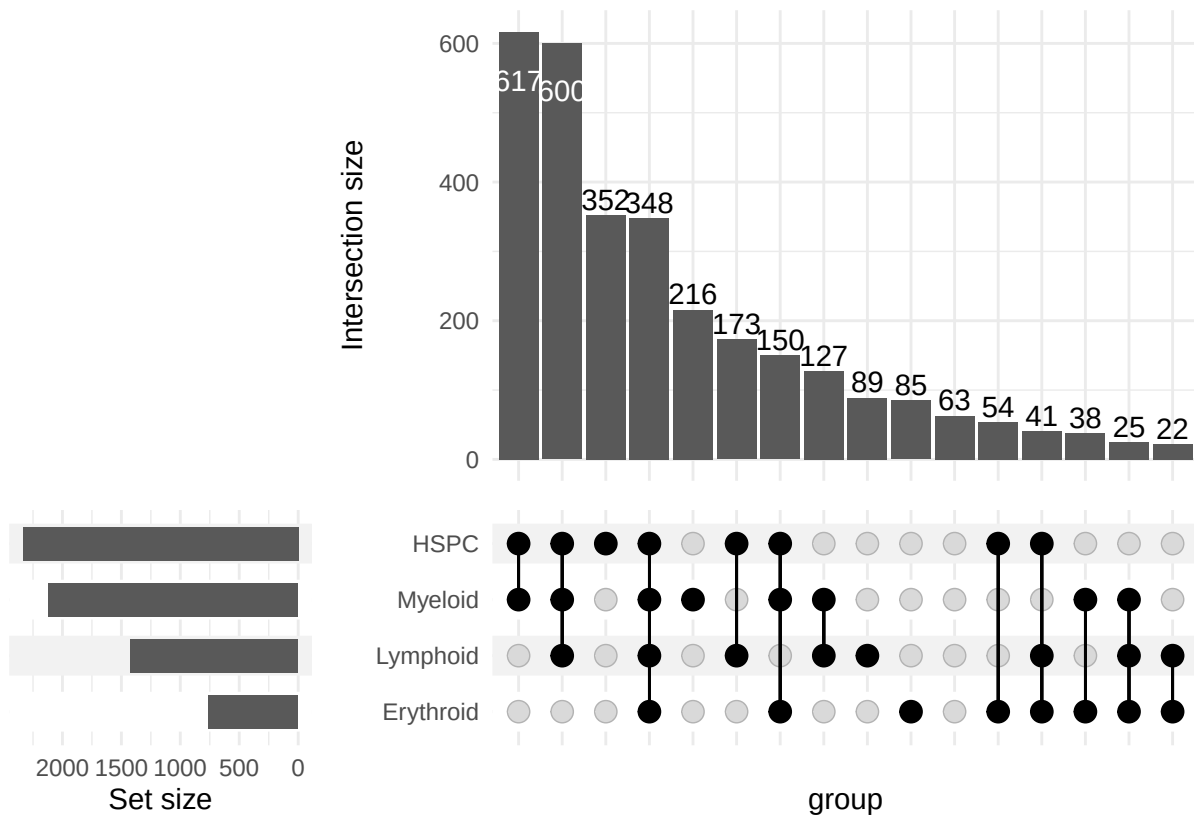
names(mutation_ct_scATA) <- samples_scATA_csm
for (sample in samples_scATA_csm){
  message(sample)
  csm_01 <- cellxmut_to_01(AML_scATA_CSM_list[[sample]])
  csm_01_ct <- celltype_matrix(mat = csm_01, meta = AML_scATA_meta[AML_scATA_meta$SampleID == sample,])
  csm_01_ct <- csm_01_ct[celltypes,]
  csm_01_ct_bin <- ifelse(csm_01_ct >= 3, 1, 0)
  df <- as.data.frame(t(csm_01_ct_bin))
  df <- df %>% rownames_to_column("mutation")
  df[celltypes] <- lapply(df[celltypes], function(x) as.logical(as.integer(x)))
  mutation_ct_scATA[[sample]] <- df
}

## AML1
## AML2
## AML3
## AML4
## AML5
## AML6
## AML8
## AML10
## AML11
## AML12
## AML13
## AML15
## AML16
## AML17
## AML18
## AML19
## AML20
## AML21
## AML22
## AML24
## AML25
## AML26
## AML27

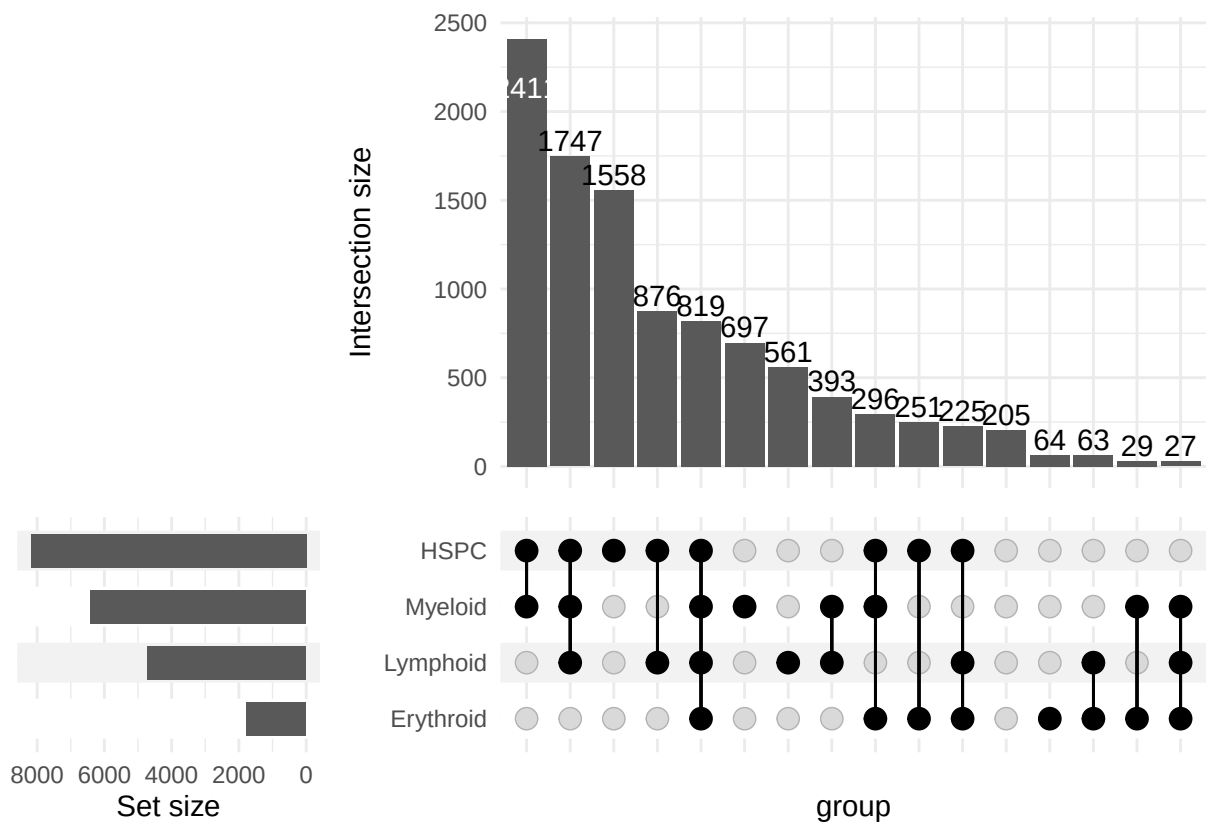
all_mutation_ct_scRNA <- do.call(rbind, mutation_ct_scRNA)
all_mutation_ct_scATA <- do.call(rbind, mutation_ct_scATA)

ComplexUpset::upset(
  all_mutation_ct_scRNA,
  intersect = celltypes,
  base_annotations = list(
    'Intersection size' = intersection_size()
  )
)

```



```
ComplexUpset::upset(
  all_mutation_ct_scATA,
  intersect = celltypes,
  base_annotations = list(
    'Intersection size' = intersection_size()
  )
)
```



```
p_all_mutation_ct_scRNA <- ComplexUpset::upset(
  all_mutation_ct_scRNA,
  intersect = celltypes,
  base_annotations = list(
    'Intersection size' = intersection_size()
  )
)
```

```
p_all_mutation_ct_scATA <- ComplexUpset::upset(
  all_mutation_ct_scATA,
  intersect = celltypes,
  base_annotations = list(
    'Intersection size' = intersection_size()
  )
)
```

```
#ggsave("../p_all_mutation_ct_scRNA.pdf", p_all_mutation_ct_scRNA, width = 4.7, height = 2.9)
#ggsave("../p_all_mutation_ct_scATA.pdf", p_all_mutation_ct_scATA, width = 4.7, height = 2.9)
```

9 SNV tree cell type

```
merge_cell_mut_matrix <- function(mat1, mat2, fill = 0) {
  all_cells <- union(rownames(mat1), rownames(mat2))
}
```

```

all_muts <- union(colnames(mat1), colnames(mat2))

merged_mat <- matrix(fill,
                     nrow = length(all_cells),
                     ncol = length(all_muts),
                     dimnames = list(all_cells, all_muts))

merged_mat[rownames(mat1), colnames(mat1)] <- mat1

merged_mat[rownames(mat2), colnames(mat2)] <-
  merged_mat[rownames(mat2), colnames(mat2)] + mat2

return(merged_mat)
}

cellxmut_to_01 <- function(cellxmut,
                          mapping = c(NoCall = 0L, WT = 0L, Mut = 1L),
                          default_value = NA_integer_,
                          return_matrix = TRUE) {
  if (!(is.matrix(cellxmut) || is.data.frame(cellxmut))) {
    stop("cellxmut must be a matrix or data.frame.")
  }

  rn <- rownames(cellxmut)
  cn <- colnames(cellxmut)
  if (is.null(rn) || any(rn == "")) stop("cellxmut must have non-empty rownames (cell names).")
  if (is.null(cn) || any(cn == "")) stop("cellxmut must have non-empty colnames (mutation sites).")

  # Work in character space, trim whitespace
  x <- as.matrix(cellxmut)
  x <- apply(x, c(1, 2), function(v) trimws(as.character(v)))

  # If already numeric-ish (e.g., "0"/"1"), keep them
  suppressWarnings(num <- matrix(as.integer(x), nrow = nrow(x), ncol = ncol(x)))
  is_num <- !is.na(num) & x %in% c("0", "1")
  out <- matrix(NA_integer_, nrow = nrow(x), ncol = ncol(x), dimnames = list(rn, cn))
  out[is_num] <- num[is_num]

  # Map remaining via mapping
  map_vec <- as.integer(mapping); names(map_vec) <- names(mapping)
  idx <- is.na(out)
  if (any(idx)) {
    mapped <- map_vec[x[idx]]
    mapped[is.na(mapped)] <- default_value
    out[idx] <- as.integer(mapped)
  }

  if (return_matrix) return(out)
  as.data.frame(out, check.names = FALSE)
}

celltype_matrix <- function(mat, meta, method = "sum") {
  common_cells <- intersect(rownames(mat), meta$CellTag)

```



```

mat <- mat[common_cells, , drop = FALSE]
meta_sub <- meta[match(common_cells, meta$CellTag), ]

df <- as.data.frame(mat)
df$Classified_CelltypeQ <- meta_sub$Classified_CelltypeQ

if (method == "sum") {
  agg <- aggregate(. ~ Classified_CelltypeQ, data = df, FUN = sum)
} else if (method == "mean") {
  agg <- aggregate(. ~ Classified_CelltypeQ, data = df, FUN = mean)
} else if (method == "binary") {
  agg <- aggregate(. ~ Classified_CelltypeQ, data = df, FUN = function(x) as.numeric(sum(x) > 0))
} else {
  stop("method must be 'sum', 'mean' or 'binary'")
}

rownames(agg) <- agg$Classified_CelltypeQ
agg$Classified_CelltypeQ <- NULL

return(as.matrix(agg))
}

celltype_matrix_state <- function(mat, meta) {

  common_cells <- intersect(rownames(mat), meta$CellTag)
  mat <- mat[common_cells, , drop = FALSE]
  ct <- meta$Classified_CelltypeQ[match(common_cells, meta$CellTag)]

  idx_list <- split(seq_along(ct), ct)

  res <- lapply(idx_list, function(idx) {

    sub_mat <- mat[idx, , drop = FALSE]

    apply(sub_mat, 2, function(x) {

      if (any(x == 1, na.rm = TRUE)) {
        return(1L)
      } else if (any(x == 0, na.rm = TRUE)) {
        return(0L)
      } else {
        return(3L)
      }

    })

  })

  res_mat <- do.call(rbind, res)

  return(res_mat)
}

```

```

AML04_CSM_scrRNA_DX <- cellxmut_to_01(cellxmut = AML_scrRNA_CSM_DX_REL_select_list$AML4_DX, mapping = c(N
AML04_CSM_scrRNA_REL <- cellxmut_to_01(cellxmut = AML_scrRNA_CSM_DX_REL_select_list$AML4_REL, mapping = c(N

AML04_CSM_scATA_DX <- cellxmut_to_01(cellxmut = AML_scATA_CSM_DX_REL_select_list$AML4_DX, mapping = c(N
AML04_CSM_scATA_REL <- cellxmut_to_01(cellxmut = AML_scATA_CSM_DX_REL_select_list$AML4_REL, mapping = c(N

AML16_CSM_scATA_DX <- cellxmut_to_01(AML_scATA_CSM_DX_REL_select_list$AML16_DX, default_value = 0L)
AML16_CSM_scATA_REL <- cellxmut_to_01(AML_scATA_CSM_DX_REL_select_list$AML16_REL, default_value = 0L)

AML18_CSM_scATA_DX <- cellxmut_to_01(AML_scATA_CSM_DX_REL_select_list$AML18_DX, default_value = 0L)
AML18_CSM_scATA_REL <- cellxmut_to_01(AML_scATA_CSM_DX_REL_select_list$AML18_REL, default_value = 0L)

AML04_CSM_scrRNA <- merge_cell_mut_matrix(AML04_CSM_scrRNA_DX,
                                          AML04_CSM_scrRNA_REL)

AML04_CSM_scATA <- merge_cell_mut_matrix(AML04_CSM_scATA_DX,
                                          AML04_CSM_scATA_REL)

AML16_CSM_scATA <- merge_cell_mut_matrix(AML16_CSM_scATA_DX,
                                          AML16_CSM_scATA_REL)

AML18_CSM_scATA <- merge_cell_mut_matrix(AML18_CSM_scATA_DX,
                                          AML18_CSM_scATA_REL)

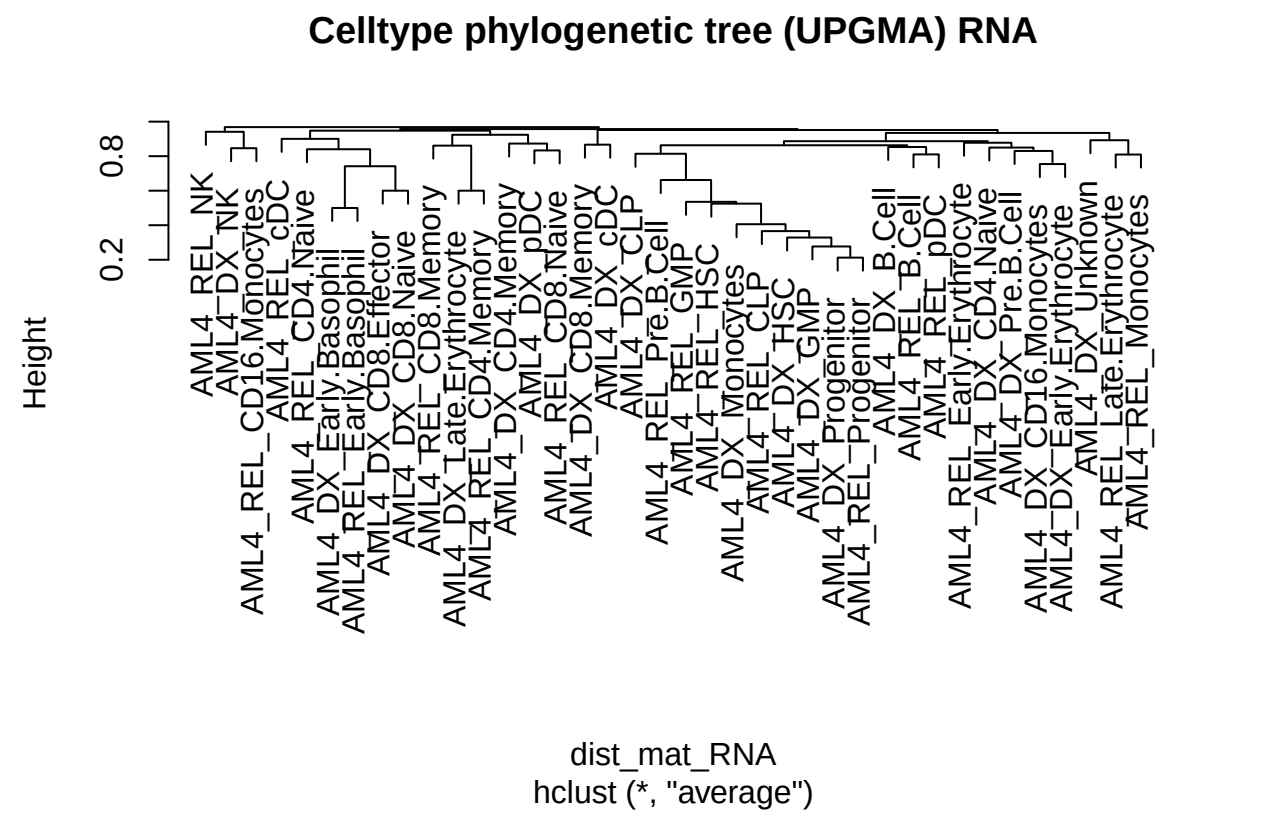
AML04_CSM_scrRNA_ct <- celltype_matrix(mat = AML04_CSM_scrRNA, meta = AML_scrRNA_meta_DX_REL)
AML04_CSM_scrRNA_ct_state <- celltype_matrix_state(AML04_CSM_scrRNA, AML_scrRNA_meta_DX_REL)
AML04_CSM_scATA_ct <- celltype_matrix(mat = AML04_CSM_scATA, meta = AML_scATA_meta_DX_REL)
AML04_CSM_scATA_ct_state <- celltype_matrix_state(AML04_CSM_scATA, AML_scATA_meta_DX_REL)
AML16_CSM_scATA_ct <- celltype_matrix(mat = AML16_CSM_scATA, meta = AML_scATA_meta_DX_REL)
AML18_CSM_scATA_ct <- celltype_matrix(mat = AML18_CSM_scATA, meta = AML_scATA_meta_DX_REL)

#write.table(x = t(AML04_CSM_scrRNA_ct_state), "/Users/liqun/Downloads/AML04_CSM_scrRNA_ct_state.txt", se
#write.table(x = t(AML04_CSM_scATA_ct_state), "/Users/liqun/Downloads/AML04_CSM_scATA_ct_state.txt", se

binary_mat_RNA <- (AML04_CSM_scrRNA_ct != 0) * 1
binary_mat_ATA <- (AML04_CSM_scATA_ct != 0) * 1
library(vegan)
library(ape)
##
## Attaching package: 'ape'
## The following object is masked from 'package:circlize':
##
## degree
## The following object is masked from 'package:ggpubr':
##
## rotate
## The following object is masked from 'package:dplyr':
##
## where
dist_mat_RNA <- vegdist(binary_mat_RNA, method = "jaccard")
dist_mat_ATA <- vegdist(binary_mat_ATA, method = "jaccard")
hc_RNA <- hclust(dist_mat_RNA, method = "average") # average = UPGMA
hc_ATA <- hclust(dist_mat_ATA, method = "average")

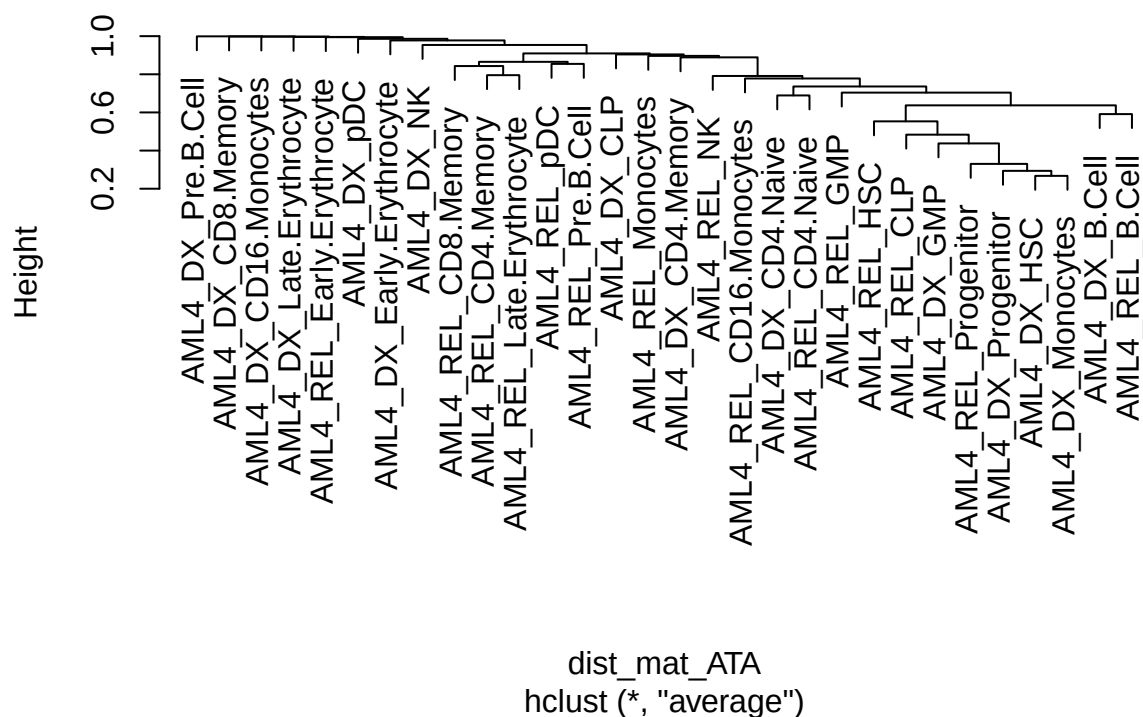
```

```
plot(hc_RNA, main = "Celltype phylogenetic tree (UPGMA) RNA")
```



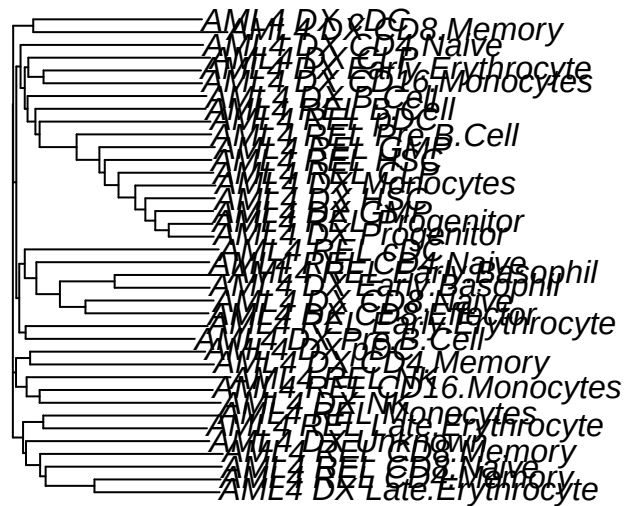
```
plot(hc_ATA, main = "Celltype phylogenetic tree (UPGMA) ATA")
```

Celltype phylogenetic tree (UPGMA) ATA



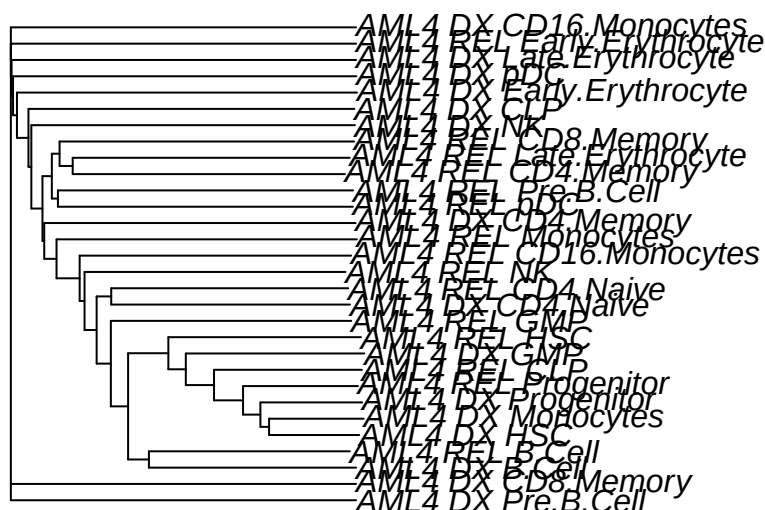
```
nj_tree_RNA <- nj(dist_mat_RNA)
nj_tree_ATA <- nj(dist_mat_ATA)
plot(nj_tree_RNA, main = "Celltype phylogenetic tree (NJ) RNA")
```

Celltype phylogenetic tree (NJ) RNA



```
plot(nj_tree_ATA, main = "Celltype phylogenetic tree (NJ) ATA")
```

Celltype phylogenetic tree (NJ) ATA



10 mutratio

```
ct_counts_scRNA <- table(
  AML_scRNA_meta_DX_REL$Classified_CelltypeQ[
    AML_scRNA_meta_DX_REL$SampleID == "AML4"
  ]
)

ct_counts_scATA <- table(
  AML_scATA_meta_DX_REL$Classified_CelltypeQ[
    AML_scATA_meta_DX_REL$SampleID == "AML4"
  ]
)

# 2. mutation counts rowSums
mut_counts_scRNA <- rowSums(AML04_CSM_scRNA_ct)
mut_counts_scATA <- rowSums(AML04_CSM_scATA_ct)

# 3. map
common_ct_scRNA <- intersect(names(ct_counts_scRNA), names(mut_counts_scRNA))
ct_counts_scRNA <- ct_counts_scRNA[common_ct_scRNA]
mut_counts_scRNA <- mut_counts_scRNA[common_ct_scRNA]
common_ct_scATA <- intersect(names(ct_counts_scATA), names(mut_counts_scATA))
ct_counts_scATA <- ct_counts_scATA[common_ct_scATA]
```

```

mut_counts_scATA <- mut_counts_scATA[common_ct_scATA]

# 4. ratio
ratio_scrNA <- mut_counts_scrNA / ct_counts_scrNA
ratio_scATA <- mut_counts_scATA / ct_counts_scATA

# 5. sort
ratio_sorted_scrNA <- sort(ratio_scrNA)
ratio_sorted_scATA <- sort(ratio_scATA)

ratio_sorted_scrNA
##
##          AML4_REL_CD4.Naive          AML4_DX_CD4.Naive
##          0.1058824          0.1288889
##          AML4_DX_CD4.Memory          AML4_REL_CD8.Naive
##          0.1395349          0.1481481
##          AML4_REL_CD4.Memory          AML4_REL_NK
##          0.1714286          0.1764706
##          AML4_REL_B.Cell          AML4_DX_B.Cell
##          0.1808511          0.1858407
##          AML4_DX_CD8.Naive          AML4_REL_CD8.Memory
##          0.2000000          0.2173913
##          AML4_DX_NK          AML4_DX_CD8.Effector
##          0.2222222          0.3333333
##          AML4_DX_Late.Erythrocyte          AML4_REL_CD16.Monocytes
##          0.3333333          0.3589744
##          AML4_REL_Monocytes          AML4_DX_CD8.Memory
##          0.3913043          0.4285714
##          AML4_REL_Late.Erythrocyte          AML4_DX_CLP
##          0.4444444          0.4460432
##          AML4_DX_CD16.Monocytes          AML4_DX_pDC
##          0.4750000          0.4761905
##          AML4_REL_Early.Basophil          AML4_DX_cDC
##          0.5000000          0.5769231
##          AML4_REL_CLP          AML4_DX_Monocytes
##          0.5908714          0.5957132
##          AML4_REL_pDC          AML4_REL_HSC
##          0.6212121          0.6294227
##          AML4_DX_Early.Basophil          AML4_DX_Pre.B.Cell
##          0.6666667          0.7187500
##          AML4_REL_Progenitor          AML4_DX_HSC
##          0.7682453          0.7790698
##          AML4_REL_cDC          AML4_REL_GMP
##          0.8000000          0.8903743
##          AML4_REL_Pre.B.Cell          AML4_DX_Early.Erythrocyte
##          0.9042553          1.0625000
##          AML4_REL_Early.Erythrocyte          AML4_DX_Progenitor
##          1.0800000          1.1514841
##          AML4_DX_GMP          AML4_DX_Unknown
##          1.2263223          1.8571429

ratio_sorted_scATA
##
##          AML4_DX_CD8.Memory          AML4_DX_Pre.B.Cell

```

```
##          0.3333333          0.3333333
## AML4_DX_Late.Erythrocyte AML4_REL_Late.Erythrocyte
##          0.5000000          0.6016260
## AML4_REL_Early.Erythrocyte AML4_REL_Pre.B.Cell
##          0.7500000          0.7727273
## AML4_REL_CD4.Memory AML4_REL_B.Cell
##          1.0126582          1.1952191
## AML4_DX_CD4.Memory AML4_REL_CD4.Naive
##          1.2000000          1.2413793
## AML4_DX_NK AML4_REL_NK
##          1.2727273          1.2893082
## AML4_DX_CD4.Naive AML4_REL_CD16.Monocytes
##          1.3400000          1.4388889
## AML4_DX_B.Cell AML4_REL_GMP
##          1.5982143          1.6376812
## AML4_REL_pDC AML4_REL_Monocytes
##          1.6521739          1.7435897
## AML4_DX_CD16.Monocytes AML4_DX_pDC
##          2.0000000          2.0000000
## AML4_DX_Monocytes AML4_REL_CLP
##          2.0700863          2.1274074
## AML4_REL_CD8.Memory AML4_REL_HSC
##          2.2000000          2.2897727
## AML4_DX_Progenitor AML4_REL_Progenitor
##          2.3260573          2.4950547
## AML4_DX_Early.Erythrocyte AML4_DX_GMP
##          2.8000000          3.0151515
## AML4_DX_HSC AML4_DX_CLP
##          3.5140845          4.6875000

common_ct <- intersect(names(ratio_scRNA), names(ratio_scATA))

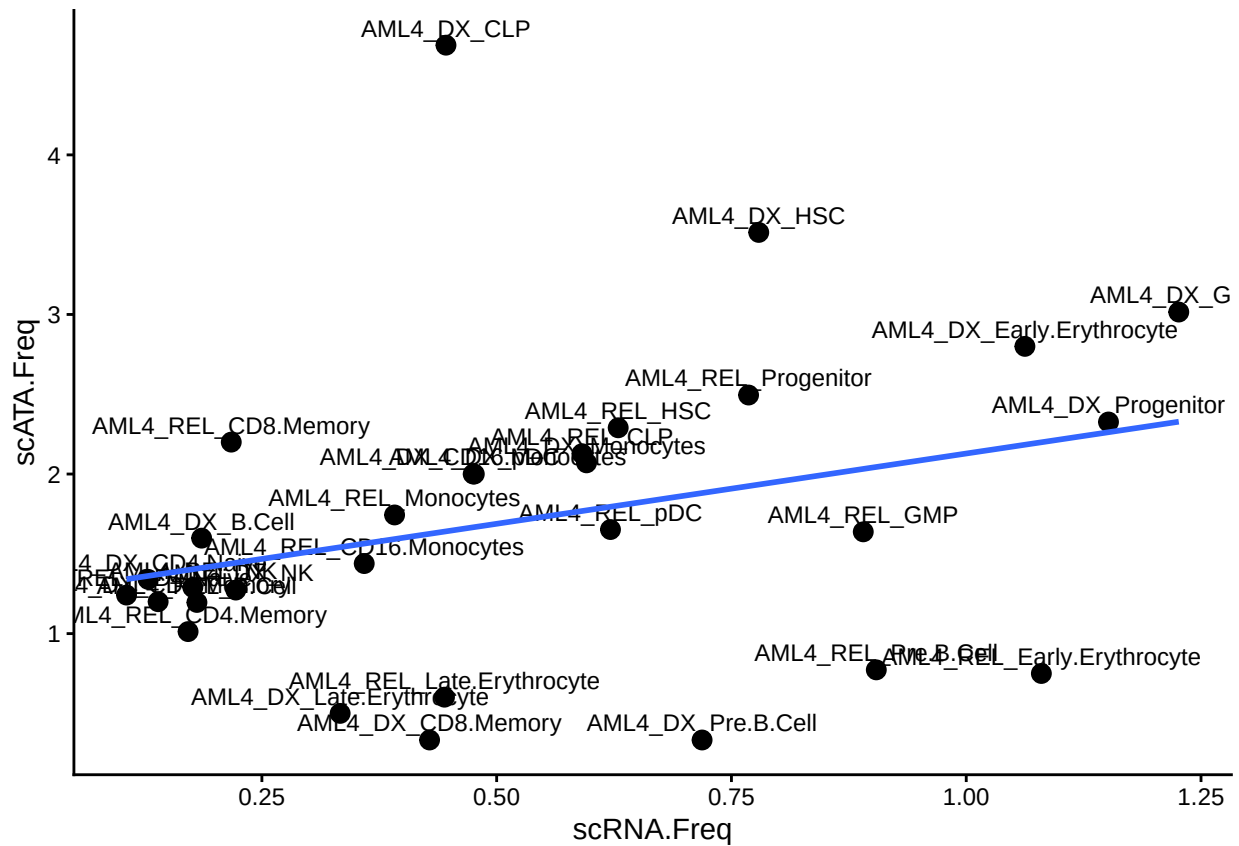
ratio_scRNA2 <- ratio_scRNA[common_ct]
ratio_scATA2 <- ratio_scATA[common_ct]
cor(ratio_scRNA2, ratio_scATA2, method = "spearman")
## [1] 0.4025367

df_cor <- data.frame(
  celltype = common_ct,
  scRNA = ratio_scRNA2,
  scATA = ratio_scATA2
)

ggplot(df_cor, aes(scRNA.Freq, scATA.Freq, label = celltype)) +
  geom_point(size = 3) +
  geom_text(size = 3, vjust = -0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  theme_classic()
## `geom_smooth()` using formula = 'y ~ x'
## Warning: The following aesthetics were dropped during statistical transformation: label.
## i This can happen when ggplot fails to infer the correct grouping structure in
## the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
```



```
## variable into a factor?
```



11 sessionInfo

```
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats4      grid        stats      graphics  grDevices  utils      datasets
## [8] methods    base
##
```

```

## other attached packages:
## [1] ape_5.8-1                        htmlwidgets_1.6.4
## [3] networkD3_0.4.1                clusterProfiler_4.16.0
## [5] ComplexHeatmap_2.24.1          org.Hs.eg.db_3.21.0
## [7] RColorBrewer_1.1-3             ggstankey_0.0.99999
## [9] ComplexUpset_1.3.6             UpSetR_1.4.0
## [11] patchwork_1.3.2                lubridate_1.9.4
## [13] forcats_1.0.0                  purrr_1.1.0
## [15] readr_2.1.5                    tibble_3.3.0
## [17] tidyverse_2.0.0                bedtoolsr_2.30.0-7
## [19] corrplot_0.95                  pheatmap_1.0.13
## [21] circlize_0.4.16                ggpmisc_0.6.2
## [23] ggpp_0.5.9                     Seurat_5.3.0
## [25] SeuratObject_5.2.0             sp_2.2-0
## [27] ggpubr_0.6.1                   tidyrr_1.3.1
## [29] GO.db_3.21.0                   AnnotationDbi_1.70.0
## [31] dplyr_1.1.4                    rhdf5_2.52.1
## [33] SummarizedExperiment_1.38.1    Biobase_2.68.0
## [35] RcppArmadillo_15.0.2-2         Rcpp_1.1.0
## [37] Matrix_1.7-4                   GenomicRanges_1.60.0
## [39] GenomeInfoDb_1.44.3            IRanges_2.42.0
## [41] S4Vectors_0.46.0              BiocGenerics_0.54.0
## [43] generics_0.1.4                 sparseMatrixStats_1.20.0
## [45] MatrixGenerics_1.20.0          matrixStats_1.5.0
## [47] data.table_1.17.8              stringr_1.5.2
## [49] plyr_1.8.9                     magrittr_2.0.4
## [51] ggplot2_4.0.0                  gtable_0.3.6
## [53] gtools_3.9.5                   gridExtra_2.3
## [55] devtools_2.4.5                 usethis_3.2.1
## [57] ArchR_1.0.3                    vegan_2.7-1
## [59] permute_0.9-8                  ineq_0.2-13
## [61] FSA_0.10.0
##
## loaded via a namespace (and not attached):
## [1] fs_1.6.6                        spatstat.sparse_3.1-0    enrichplot_1.28.4
## [4] doParallel_1.0.17              httr_1.4.7              profvis_0.4.0
## [7] tools_4.5.1                    sctransform_0.4.2       backports_1.5.0
## [10] R6_2.6.1                       lazyeval_0.2.2          uwot_0.2.3
## [13] mgcv_1.9-3                     GetoptLong_1.0.5        rhdf5filters_1.20.0
## [16] urlchecker_1.0.1              withr_3.0.2             progressr_0.16.0
## [19] quantreg_6.1                   cli_3.6.5               Cairo_1.6-5
## [22] spatstat.explore_3.5-3         fastDummies_1.7.5       labeling_0.4.3
## [25] S7_0.2.0                       spatstat.data_3.1-8     ggribes_0.5.7
## [28] pbapply_1.7-4                  yulab.utils_0.2.1       gson_0.1.0
## [31] R.utils_2.13.0                 DOSE_4.2.0              dichromat_2.0-0.1
## [34] parallelly_1.45.1             sessioninfo_1.2.3       rstudioapi_0.17.1
## [37] RSQLite_2.4.3                  gridGraphics_0.5-1      shape_1.4.6.1
## [40] ica_1.0-3                      spatstat.random_3.4-2   car_3.1-3
## [43] abind_1.4-8                    R.methodsS3_1.8.2       lifecycle_1.0.4
## [46] yaml_2.3.10                    carData_3.0-5           qvalue_2.40.0
## [49] SparseArray_1.8.1             Rtsne_0.17              blob_1.2.4
## [52] promises_1.3.3                 crayon_1.5.3            ggtangle_0.0.7
## [55] miniUI_0.1.2                   lattice_0.22-7          cowplot_1.2.0

```

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## [58] KEGGREST_1.48.1      pillar_1.11.1      knitr_1.50
## [61] fgsea_1.34.2         rjson_0.2.23       future.apply_1.20.0
## [64] codetools_0.2-20     fastmatch_1.1-6    glue_1.8.0
## [67] ggfun_0.2.0          spatstat.univar_3.1-4 remotes_2.5.0
## [70] treeio_1.32.0        vctrs_0.6.5        png_0.1-8
## [73] spam_2.11-1          cachem_1.1.0       xfun_0.53
## [76] S4Arrays_1.8.1       mime_0.13          survival_3.8-3
## [79] iterators_1.0.14     ellipsis_0.3.2     fitdistrplus_1.2-4
## [82] ROCR_1.0-11          nlme_3.1-168       ggtree_3.16.3
## [85] bit64_4.6.0-1        RcppAnnoy_0.0.22   data.tree_1.2.0
## [88] irlba_2.3.5.1        KernSmooth_2.23-26 colorspace_2.1-2
## [91] DBI_1.2.3            tidyselect_1.2.1   bit_4.6.0
## [94] compiler_4.5.1       SparseM_1.84-2     DelayedArray_0.34.1
## [97] plotly_4.11.0        scales_1.4.0       lmtest_0.9-40
## [100] rappdirs_0.3.3       digest_0.6.37      goftest_1.2-3
## [103] spatstat.utils_3.2-0 rmarkdown_2.30     XVector_0.48.0
## [106] htmltools_0.5.8.1    pkgconfig_2.0.3    fastmap_1.2.0
## [109] rlang_1.1.6          GlobalOptions_0.1.2 UCSC.utils_1.4.0
## [112] shiny_1.11.1         farver_2.1.2       zoo_1.8-14
## [115] jsonlite_2.0.0       BiocParallel_1.42.2 R.oo_1.27.1
## [118] GOSemSim_2.34.0      ggplotify_0.1.3    polynom_1.4-1
## [121] Formula_1.2-5        GenomeInfoDbData_1.2.14 dotCall64_1.2
## [124] Rhdf5lib_1.30.0      reticulate_1.43.0  stringi_1.8.7
## [127] MASS_7.3-65          pkgbuild_1.4.8     parallel_4.5.1
## [130] listenv_0.9.1        ggrepel_0.9.6      deldir_2.0-4
## [133] Biostrings_2.76.0    splines_4.5.1      tensor_1.5.1
## [136] hms_1.1.3            igraph_2.1.4       spatstat.geom_3.6-0
## [139] ggsignif_0.6.4       RcppHNSW_0.6.0     reshape2_1.4.4
## [142] pkgload_1.4.0        evaluate_1.0.5     foreach_1.5.2
## [145] tzdb_0.5.0           httpuv_1.6.16      MatrixModels_0.5-4
## [148] RANN_2.6.2           polyclip_1.10-7    clue_0.3-66
## [151] future_1.67.0        scattermore_1.2    broom_1.0.10
## [154] xtable_1.8-4         tidytree_0.4.6     RSpectra_0.16-2
## [157] rstatix_0.7.2        later_1.4.4        viridisLite_0.4.2
## [160] aplot_0.2.9          memoise_2.0.1      cluster_2.1.8.1
## [163] timechange_0.3.0     globals_0.18.0

```